INTRODUCTION TO STOCHASTIC PROCESSES

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Fifth Semester

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Chapter 1

RANDOM WALKS

July 22nd.

Quizzes will account for 20% of the final grade, and assignments for 5%. The mid-term will account for 25% of the final grade, and the final exam will account for 50%. The primary reference for random walks is H. Follner.

1.1 The Setup

We wish to concretify the idea of a random walk on the integers $\mathbb{Z}$. Ω is out sample space, with $0 < |\Omega| < \infty$ (we shall later look at the case when Ω is countably infinite). $\mathcal{F}$ is the event space, and is exactly $\mathcal{P}(\Omega)$, the power set of Ω. The intuition of a random walk is as follows; stand at the origin. At each step, flip a fair coin. If the coin comes up heads, take a step to the right; if tails, take a step to the left. Repeat this process N times. S_n here then denotes the position of the random walk after n steps, for $0 \leq n \leq N$.

For a fixed $N \in \mathbb{N}$, we let $\Omega_N = \{\pm 1\}^N$, and $\mathcal{F}_N = \mathcal{P}(\Omega_N)$. For $1 \leq k \leq N$, define the random variable $X_k : \Omega_N \to \{\pm 1\}$ as $X_k(\omega) = \omega_k$, where $\omega = (\omega_1, \omega_2, \ldots, \omega_N) \in \Omega_N$; this models the step taken by the random walk at step k. Further, define $S_0 = 0$, and for $1 \leq n \leq N$, define $S_n : \Omega_N \to \mathbb{Z}$ as $S_n(\omega) = \sum_{k=1}^{n} X_k(\omega)$; this models the position of the random walk after n steps. Finally, define the probability measure $\mathbb{P}_N$ on $(\Omega_N, \mathcal{F}_N)$ as $\mathbb{P}_N(A) = \frac{|A|}{2^N}$ for $A \in \mathcal{F}_N$. This models the fair coin flip at each step of the random walk.

Definition 1.1. The sequence of random variables $\{S_n\}_{n=0}^N$ on $(\Omega_N, \mathcal{F}_N, \mathbb{P}_N \equiv \mathbb{P})$, as prescribed above, is called the *simple symmetric random walk* of length N on $\mathbb{Z}$, starting at the origin.

Given $1 \leq k_1 < k_2 < \cdots < k_n \leq N$, and $x_i \in \{\pm 1\}$,

$$\mathbb{P}(X_{k_1} = x_1, \ldots, X_{k_n} = x_n) = \frac{|\{\omega \in \Omega_N \mid X_{k_i}(\omega) = x_i, 1 \leq i \leq n\}|}{2^N} = \frac{2^{N-n}}{2^N} = 2^{-n}. \tag{1.1}$$

That is, fixing n steps of the walk, the probability of any particular sequence of steps is 2^{-n}. Using this, we can show that for any $1 \leq k \leq N$, $\mathbb{P}(X_k = 1) = \mathbb{P}(X_k = -1) = \frac{1}{2}$. Moreover, we can extract that

$$\mathbb{P}(X_k = x_k, X_l = x_l) = 2^{-2} = \mathbb{P}(X_k = x_k)\mathbb{P}(X_l = x_l), \tag{1.2}$$

showing independence of subsequent steps; the model seems like a reasonable fit for our intuition of a random walk.

Example 1.2. The simple symmetric random walk has independent increments; that is, given $0 < k_1 < k_2 < N$, the random variables $S_k - S_0$, $S_{k_2} - S_{k_1}$, and $S_N - S_{k_2}$ are mutually independent. This is because the increments are functions of disjoint sets of independent random variables X_k, and hence are independent themselves. Similarly, one may concretely show that for $1 \leq k < m \leq N$, the distribution of $S_m - S_k$ is the same as that of S_{m-k}.

The following properties of the simple symmetric random walk are immediate consequences of the above definition and example.

Suppose we have $\mathbb{P}(S_{n-1} = a_{n-1}, S_{n-2} = a_{n-2}, \ldots, S_0 = 0) > 0$. Then, we are free to compute the conditional probability $\mathbb{P}(S_n = a_n \mid S_{n-1} = a_{n-1}, S_{n-2} = a_{n-2}, \ldots, S_0 = 0)$. One may verify that

$$\mathbb{P}(S_n = a_n \mid S_{n-1} = a_{n-1}, S_{n-2} = a_{n-2}, \ldots, S_0 = 0) = \mathbb{P}(S_n = a_n \mid S_{n-1} = a_{n-1}), \tag{1.3}$$

for all $a_1, \ldots, a_n \in \mathbb{Z}$ such that $\mathbb{P}(S_{n-1} = a_{n-1}, S_{n-2} = a_{n-2}, \ldots, S_0 = 0) > 0$. This is called the *Markov property* of the simple symmetric random walk. In words, this property states that given the present state of the walk, the future evolution of the walk is independent of its past history.

Now suppose $\mathbb{P}(S_k = a) > 0$ for some $k \geq 1, a \in \mathbb{Z}$, and let $m > k, b \in \mathbb{Z}$. Then, one can simply compute that $\mathbb{P}(S_m = b \mid S_k = a) = \mathbb{P}(S_{m-k} = b - a)$.

The next question we may ask is, on an average, where the random walk is located after n steps, for $1 \leq n \leq N$. We can compute the expected value of S_n as follows:

$$\mathbb{E}[S_n] = \mathbb{E}\left[\sum_{k=1}^{n} X_k + 0\right] = \sum_{k=1}^{n} \mathbb{E}[X_k] = \sum_{k=1}^{n} 0 = 0. \tag{1.4}$$

Thus, to get a better idea of the location of the random walk after n steps, we compute the variance of S_n:

$$\mathrm{Var}(S_n) = \mathrm{Var}\left(\sum_{k=1}^{n} X_k + 0\right) = \sum_{k=1}^{n} \mathrm{Var}(X_k) = \sum_{k=1}^{n} 1 = n. \tag{1.5}$$

So, via the central limit theorem, we can approximate for large n as

$$\mathbb{P}\left(-3 \leq \frac{S_n}{\sqrt{n}} \leq 3\right) \approx \Phi(3) - \Phi(-3) \tag{1.6}$$

where Φ is the cumulative distribution function of the standard normal distribution.

We now claim that

$$\mathbb{P}(S_{2n} = 0) = \binom{2n}{n} 2^{-2n} = \mathbb{P}(S_{2n-1} = 1) = \mathbb{P}(S_{2n-1} = -1). \tag{1.7}$$

To this end, we note that for $S_n = x$, we must have taken $\frac{n+x}{2}$ steps to the right, and $\frac{n-x}{2}$ steps to the left. Thus, we have

$$\mathbb{P}(S_n = x) = \frac{|\{\omega \in \Omega_N \mid S_n = x\}|}{2^N} = \binom{n}{\frac{n+x}{2}} \frac{2^{N-n}}{2^N} = \binom{n}{\frac{n+x}{2}} 2^{-n}. \tag{1.8}$$

The binomial coefficient attains its maximum at $x = 0$ for even n, and at $x = \pm 1$ for odd n. Thus, we have shown the above required claim. We can further approximate this via *Stirling's formula*, which stats $k! \sim \sqrt{2\pi k} k^k e^{-k}$ for large k. For now, we shall treat this as an equality. Thus, for large n,

$$\mathbb{P}(S_{2n} = 0) = \binom{2n}{n} 2^{-2n} \sim \frac{\sqrt{4\pi n}(2n)^{2n} e^{-2n}}{2\pi n n^{2n} e^{-2n}} 2^{-2n} = \frac{1}{\sqrt{\pi n}}. \tag{1.9}$$

We now show that given any interval $[a, b] \subsetneq \mathbb{R}$, the probability that the random walk stays in this interval is zero as $n \to \infty$. To this end, note that either $\mathbb{P}(S_n = x) \leq \mathbb{P}(S_n = 0)$ or $\mathbb{P}(S_n = x) \leq \mathbb{P}(S_n = 1)$ holds for any given n; moreover, both of the quantities on the right-hand side are bounded by $\frac{c}{\sqrt{n}}$ for an appropriate constant $c > 0$. Thus, we have

$$0 \leq \mathbb{P}(a \leq S_n \leq b) = \sum_{x=a}^{b} \mathbb{P}(S_n = x) \leq \frac{c(b - a + 1)}{\sqrt{n}} \to 0 \text{ as } n \to \infty. \tag{1.10}$$

1.2 Stopping Times

July 29th.

We can interpret S_n as the amount of capital earned by a player until time n. Earlier, we saw that the expected 'amount of capital' at n is $\mathbb{E}[S_n] = 0$, for $0 \leq n \leq N$. We ask if there is a favourite (random) time at which the player should stop playing, so as to maximize the expected amount of capital earned. We assume that there is no insider trading, and that the player's decision to stop playing at time n is based on the information available until time n.

Definition 1.3. A subset $A \subseteq \Omega$ is termed an *observable event* by time n if it can be written as a union of basic events of the form

$$\{\omega \in \Omega \mid \omega_1 = o_1, \ldots, \omega_n = o_n\}, \quad o_1, \ldots, o_n \in \{\pm 1\}. \tag{1.11}$$

That is, A can be determined by observing the outcomes of the first n coin tosses.

We shall use $\mathcal{A}_n$ to denote the collection of all observable events by time n. One can show that

$$\{\emptyset, \Omega\} = \mathcal{A}_0 \subseteq \mathcal{A}_1 \subseteq \cdots \subseteq \mathcal{A}_N = \mathcal{F}. \tag{1.12}$$

Definition 1.4. A map $T : \Omega \to \{0, 1, \ldots, N\} \cup \{\infty\}$ is called a *stopping time* if $\{T = n\} \in \mathcal{A}_n$ for all $0 \leq n \leq N$.

As an example, pick $a \in \mathbb{Z}$. We can then define $\sigma_a : \Omega \to \{0, 1, 2, \ldots, N\} \cup \{\infty\}$ as $\sigma_a(\omega) = \min\{n \geq 0 \mid S_n(\omega) = a\}$, with the convention that $\min \emptyset = \infty$. This is, simply put, the first time the random walk hits the point a. The reader may show that σ_a is indeed a stopping time. On the other hand, the map $\tau : \Omega \to \{0, 1, 2, \ldots, N\} \cup \{\infty\}$ defined as $\tau(\omega) = \min\{n \geq 0 \mid S_n(\omega) = S_N(\omega)\}$ is not a stopping time; this is because we cannot determine if the random walk has hit its final position at time n without knowing the outcome of the coin tosses after time n.

Theorem 1.5. *For any stopping time $T : \Omega \to \{0, 1, \ldots, N\}$, we have $\mathbb{E}[S_T] = 0$.*

That is, there is an impossibility of a favourite stopping time. We shall adopt the notation $S_T(\omega) := S_{T(\omega)}(\omega)$, the outcome of the trajectory ω at the stopping time $T(\omega)$. This leads us to discuss the impossibillity of profitable games.

Let $V_1, \ldots, V_N : \Omega \to \mathbb{R}$ be random variables such that $\{V_k = c\} \in \mathcal{A}_{k-1}$ for all $c \in \mathbb{R}$, $k = 1, \ldots, N$. V_k here represents the amount of capital a player bets at time k. The result at the k-th round of the game is then $V_k X_k$. This is known as a game system.

Theorem 1.6. *For any $V_1, \ldots, V_N$ which are a game system as above, let $S_n^V = \sum_{k=1}^n V_k X_k$ for $1 \leq n \leq N$. Then, $\mathbb{E}[S_N^V] = 0$.*

Proof. We have $\mathbb{E}[S_N^V] = \sum_{k=1}^N \mathbb{E}[V_k X_k]$. Let us restrict our range as $\text{range}(V_k) = \{c_1, \ldots, c_M\}$ for some $M \in \mathbb{N}$. Then, we have

$$\mathbb{E}[V_k X_k] = \sum_{i=1}^M c_i \, \mathbb{E}[\mathbf{1}_{\{V_k = c_i\}} X_k] \tag{1.13}$$

Note that, by definition, $\{V_k = c_i\} \in \mathcal{A}_{k-1}$, and hence is independent of X_k. Thus, we have

$$\mathbb{E}[V_k X_k] = \sum_{i=1}^M c_i \, \mathbb{E}[\mathbf{1}_{\{V_k = c_i\}}] \, \mathbb{E}[X_k] = 0. \tag{1.14}$$

■

We shall use this theorem to prove the theorem before it.

Proof of impossibility of a favourite stopping time. We have

$$S_T = \sum_{k=1}^{T(\omega)} X_k = \sum_{k=1}^N \mathbf{1}_{\{T \geq k\}} X_k. \tag{1.15}$$

Now, for $1 \leq k \leq N$, set $V_k : \Omega \to \{0, 1\}$ as $V_k(\omega) = \mathbf{1}_{\{T(\omega) \geq k\}}$. Then $\{V_k = 1\} = \{T \geq k\} = \{T < k\}^c = \left(\bigcup_{m=0}^{k-1} \{T = m\}\right)^c \in \mathcal{A}_{k-1}$ since $\{T = m\} \in \mathcal{A}_m \subseteq \mathcal{A}_{k-1}$ for $0 \leq m \leq k-1$, and $c\mathcal{A}_{k-1}$ is closed under finite unions and complements. Thus, $V_1, \ldots, V_N$ is a game system, and hence $\mathbb{E}[S_T] = \mathbb{E}[S_N^V] = 0$ by the previous theorem. ■

Suppose $T : \Omega \to \{0, 1, \ldots, N\}$ is a stopping time. Then $\mathrm{Var}(S_T) = \mathbb{E}[S_T^2] - \mathbb{E}[S_T]^2 = \mathbb{E}[S_T^2]$. We can compute $\mathbb{E}[S_T^2]$ as follows; let $1 \leq k \leq N$. Then

$$S_k^2 = (S_{k-1} + X_k)^2 = S_{k-1}^2 + 2X_k S_{k-1} + 1. \tag{1.16}$$

Set $V_k = S_{k-1}\mathbf{1}_{\{T \geq k\}}$. Verify that $V_1, \ldots, V_N$ is a game system. Then, we have

$$\begin{aligned} S_N^V &= \sum_{k=1}^N X_k V_k = \sum_{k=1}^N (S_{k-1}X_k)\mathbf{1}_{\{T\geq k\}} = \sum_{k=1}^N \frac{1}{2}(S_k^2 - S_{k-1}^2 - 1)\mathbf{1}_{\{T\geq k\}} \\ &= \frac{1}{2}\sum_{k=1}^N S_k^2 \mathbf{1}_{\{T\geq k\}} - \frac{1}{2}\sum_{k=1}^N S_{k-1}^2 \mathbf{1}_{\{T\geq k\}} - \frac{1}{2}\sum_{k=1}^N \mathbf{1}_{\{T\geq k\}} \\ &= \frac{1}{2}\sum_{k=1}^N S_k^2 \mathbf{1}_{\{T=k\}} - \frac{T}{2} = \frac{S_T^2 - T}{2}. \end{aligned} \tag{1.17}$$

By the previous theorem, we have $\mathbb{E}[S_N^V] = 0$, and hence $\mathbb{E}[S_T^2] = \mathbb{E}[T]$.

1.2.1 Ruin Problem

We have two players, one with capital a and the other with capital b. For $0 \leq n \leq N$, let S_n denote the gain of the first player. We define the first player's ruin as

$$\{\sigma_{-a} < \sigma_b,\, \sigma_{-a} \leq N\}. \tag{1.18}$$

Define the ruin probability γ_N^1 to be the probability of this event, the first player's ruin. Similarly, we define the second player's ruin as

$$\{\sigma_b < \sigma_{-a},\, \sigma_b \leq N\}, \tag{1.19}$$

and define the ruin probability γ_N^2 to be the probability of this event, the second player's ruin. Now define $T_N : \Omega \to \{0, 1, 2, \ldots, N\}$ as $T_N = \min\{\sigma_{-a}, \sigma_b, N\}$. Then, T_N is a stopping time. It has been shown that $\mathbb{E}[S_{T_N}] = 0$. Explicitly writing out S_{T_N}, we have

$$S_{T_N} = \begin{cases} -a & \text{if } \sigma_{-a} < \sigma_b \text{ and } \sigma_{-a} \leq N, \\ b & \text{if } \sigma_b < \sigma_{-a} \text{ and } \sigma_b \leq N, \\ S_N & \text{if } \min(\sigma_{-a}, \sigma_b) \geq N. \end{cases} \tag{1.20}$$

Taking expectations, we have

$$\begin{aligned} 0 &= -a\mathbb{P}(\sigma_{-a} < \sigma_b,\, \sigma_{-a} \leq N) + b\mathbb{P}(\sigma_b < \sigma_{-a},\, \sigma_b \leq N) + \mathbb{E}[S_N \mathbf{1}_{\{\min(\sigma_{-a},\sigma_b)\geq N\}}] \\ \implies 0 &= -a\gamma_N^1 + b\gamma_N^2 + \mathbb{E}[S_N \mathbf{1}_{\{\min(\sigma_{-a},\sigma_b)\geq N\}}]. \end{aligned} \tag{1.21}$$

Here, we claim that $\mathbb{E}[S_N \mathbf{1}_{\{\min(\sigma_{-a},\sigma_b)\geq N\}}] \to 0$ as $N \to \infty$. To this end, first notice the bounds

$$0 \leq |S_n|\, \mathbf{1}_{\{\min(\sigma_{-a},\sigma_b)\geq N\}} \leq (b+a)\mathbf{1}_{\{\min(\sigma_{-a},\sigma_b)\geq N\}} \tag{1.22}$$

which then implies

$$0 \leq \mathbb{E}[|S_n|\, \mathbf{1}_{\{\min(\sigma_{-a},\sigma_b)\geq N\}}] \leq (b+a)\mathbb{P}(\min(\sigma_{-a},\sigma_b) \geq N) \leq (b+a)c\frac{(b-a+1)}{\sqrt{N}} \to 0 \text{ as } N \to \infty. \tag{1.23}$$

We now make three 'leaps of faith;' they are

1. $\gamma^1 = \lim_{N\to\infty} \gamma_N^1$ and $\gamma^2 = \lim_{N\to\infty} \gamma_N^2$ exist.
2. $\gamma^1 + \gamma^2 = 1$.
3. $0 = -a\gamma^1 + b\gamma^2$.

The last two assumptions imply that $\gamma^1 = \frac{b}{a+b}$ and $\gamma^2 = \frac{a}{a+b}$.

1.3 Reflection Principle

Consider a random walk starting at the origin that ends up at $a+c$. Such a walk will have to pass through a at some point. We can then reflect the part of the walk after it hits a about the line $y = a$. This gives us a bijection between walks that end up at $a + c$ and walks that reach a and then end up at $a - c$. That is,

$$|\{\omega \in \Omega \mid S_n = a + c\}| = |\{\omega \in \Omega \mid S_n = a + c,\, \sigma_a \le n\}| = |\{\omega \in \Omega \mid S_n = a - c,\, \sigma_a \le n\}|. \tag{1.24}$$

This lemma immediately follows.

Lemma 1.7. $\mathbb{P}(S_n = a + c) = \mathbb{P}(\sigma_a \le n,\, S_n = a - c)$.

Theorem 1.8. $\mathbb{P}(\sigma_a \le n) = \mathbb{P}(S_n \notin [-a, a-1])$, *and* $\mathbb{P}(\sigma_a = n) = \frac{a}{n}\mathbb{P}(S_n = a)$.

Proof. We have

$$\begin{aligned}\mathbb{P}(\sigma_a \le n) &= \sum_{b\in\mathbb{Z}} \mathbb{P}(\sigma_a \le n,\, S_n = b) = \sum_{b\in\mathbb{Z},\, b\ge a} \mathbb{P}(\sigma_a \le n,\, S_n = b) + \sum_{b\in\mathbb{Z},\, b<a} \mathbb{P}(\sigma_a \le n,\, S_n = b)\\ &= \mathbb{P}(S_n \ge a) + \sum_{b\in\mathbb{Z},\, b<a} \mathbb{P}(S_n = 2a - b) = \mathbb{P}(S_n \ge a) + \mathbb{P}(S_n > a) = \mathbb{P}(S_n \ge a) + \mathbb{P}(S_n < -a)\\ &= \mathbb{P}(S_n \notin [-a, a-1]).\end{aligned} \tag{1.25}$$

On the other hand,

$$\begin{aligned}\mathbb{P}(\sigma_a = n) &= \mathbb{P}(\sigma_a \le n) - \mathbb{P}(\sigma_a \le n-1) = \frac{1}{2}\left(\mathbb{P}(S_{n-1} = a-1) - \mathbb{P}(S_{n-1} = a+1)\right)\\ &= \frac{1}{2}\left(\binom{n-1}{\frac{n-1+a-1}{2}}\frac{1}{2^{n-1}} - \binom{n-1}{\frac{n-1-a+1}{2}}\frac{1}{2^{n-1}}\right) = \frac{1}{2^n}\cdot\frac{a}{n}\binom{n}{\frac{n+a}{2}} = \frac{a}{n}\mathbb{P}(S_n = a).\end{aligned} \tag{1.26}$$

■

We now take as convention that $\sigma_0 = \min\{n \ge 1 \mid S_n = 0\}$ so that $\sigma_0 \ne 0$.

Lemma 1.9. $\mathbb{P}(\sigma_0 > 2n) = \mathbb{P}(S_{2n-1} = -1)$.

Proof. We have

$$\begin{aligned}\mathbb{P}(\sigma_0 > 2n) &= \mathbb{P}(S_1 \ne 0, S_2 \ne 0, \dots, S_{2n} \ne 0) = 2\mathbb{P}(S_1 > 0, \dots, S_{2n} > 0)\\ &= 2\cdot\frac{1}{2^{2n}}\#\{\text{paths that start at 0 and don't hit } -1 \text{ in } 2n-1 \text{ steps}\}\\ &= \frac{1}{2^{2n-1}}\#\{\text{paths that start at 0 and stay below 1 in } 2n-1 \text{ steps}\} = \mathbb{P}(\sigma_1 > 2n-1)\\ &= 1 - \mathbb{P}(\sigma_1 \le 2n-1) = 1 - \mathbb{P}(S_{2n-1} \notin [0,1]) = 1 - \mathbb{P}(S_{2n-1} \ne -1) = \mathbb{P}(S_{2n-1} = -1).\end{aligned} \tag{1.27}$$

■

1.3.1 Arcsine Law

August 5th.

The arcsine law is concerned with the analysis of the last visit to the origin till $2N$. That is, defining $L = \max\{0 \le n \le 2N \mid S_n = 0\}$, we wish to compute the order of L.

Theorem 1.10. *Let* $n \in \mathbb{N}_0$, *with* $n \le N$. *Then* $\mathbb{P}(L = 2n) = \frac{1}{2^{2N}}\binom{2n}{n}\binom{2N-2n}{N-n}$.

Remark 1.11. Suppose this theorem holds; using Stirling's formula, that $n! \sim n^n e^{-n}$ for large n, we can approximate $\mathbb{P}(L = 2n) \sim \frac{1}{\pi\sqrt{n(N-n)}} = \frac{1}{\pi N}\frac{1}{\sqrt{\frac{n}{N}(1-\frac{n}{N})}}$. The idea is to find the order of $\mathbb{P}(L \le 2Nx) = \mathbb{P}(\frac{L}{2N} \le x)$ for $x \in [0,1]$. Rewriting gives

$$\mathbb{P}(L \le 2Nx) = \sum_{n \le \lfloor Nx \rfloor} \mathbb{P}(L = 2n) = \sum_{n \le \lfloor Nx \rfloor} \frac{1}{N} f(\frac{1}{N}) \tag{1.28}$$

where $f(x) = \frac{1}{\pi\sqrt{x(1-x)}}$. This is a Riemann sum, and hence we have $\mathbb{P}(L \le 2Nx) \sim \int_0^x f(t)dt = \frac{2}{\pi}\arcsin(\sqrt{x})$.

Proof of theorem. We simply have

$$\mathbb{P}(L = 2n) = \mathbb{P}(S_{2n} = 0)\mathbb{P}(\sigma_0 > 2N - 2n) = \binom{2n}{n}\frac{1}{2^{2n}} \cdot \mathbb{P}(S_{2N-2n} = 0) = \frac{1}{2^{2N}}\binom{2n}{n}\binom{2N-2n}{N-n}. \tag{1.29}$$

■

1.4 Higher Dimensions and Infinite Walks

We are now fit to make sense of simple random walks on higher dimensions; let $\mathbb{Z}^d = \{(x_1, \dots, x_d) \mid x_i \in \mathbb{Z}, 1 \le i \le d\}$ for $d \ge 1$. Also, let $|x| = \sqrt{\sum_{i=1}^d x_i^2}$. Then, for a fixed $N \ge 1$, we define

$$\Omega_N = \{\omega = (\omega_1, \dots, \omega_N) \mid \omega_i \in \mathbb{Z}^d,\ |\omega_i| = 1,\ 1 \le i \le N\} \tag{1.30}$$

and $\xi_i : \Omega_N \to \mathbb{Z}^d$ as $\xi_i(\omega) = \omega_i$ for $1 \le i \le N$. We can then further define $X_n : \Omega_N \to \mathbb{Z}^d$ as $X_n(\omega) = \sum_{i=1}^n \xi_i(\omega)$ for $1 \le n \le N$. Moreover, $\mathcal{F} = \mathcal{P}(\Omega_N)$, and $\mathbb{P}(A) = \frac{|A|}{(2d)^N}$ for $A \in \mathcal{F}$. This is the simple symmetric random walk on $\mathbb{Z}^d$ of length N.

We wish to extend our notion of random walks for ones of infinite length; that is, the consistency of $\mathbb{P}_N$ as N varies. We define the projection map $\Pi_M : \Omega_N \to \Omega_M$ as $\omega \mapsto (\omega_1, \dots, \omega_M) =: \tilde{\omega}$. Now fix $\tilde{\omega} \in \Omega_M$ so that

$$\mathbb{P}_N(\{\omega \in \Omega_N \mid \Pi_M(\omega) = \tilde{\omega}\}) = \frac{|\{\omega \in \Omega_N \mid \Pi_N(\omega) = \tilde{\omega}\}|}{(2d)^N} = \frac{(2d)^{N-M}}{(2d)^N} = (2d)^{-M} = \mathbb{P}_M(\{\tilde{\omega}\}). \tag{1.31}$$

Theorem 1.12 (*Kolmogorov consistency theorem*)**.** *There exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that $\Omega \equiv \Omega_\infty = \{(\omega_i)_{i \ge 1} \mid \omega_i \in \mathbb{Z}^d,\ |w_i| = 1\}$, with $\mathcal{F}$ suitably defined, and $\mathbb{P} : \mathcal{F} \to [0,1]$ such that $\mathbb{P}(\omega \in \Omega \mid \Pi_M(\omega) = \tilde{\omega}) = \mathbb{P}_M(\{\tilde{\omega}\})$ for all $M \ge 1$ and $\tilde{\omega} \in \Omega_M$.*

A proof of this theorem is beyond the scope of this course; this lays the groundwork for the study of random walks of infinite length, by defining $\xi_i : \Omega \to \mathbb{Z}^d$ as $\xi_i(\omega) = \omega_i$ for $i \ge 1$, and $X_n : \Omega \to \mathbb{Z}^d$ as $X_n(\omega) = \sum_{i=1}^n \xi_i(\omega)$ for $n \ge 1$. This is the simple symmetric random walk on $\mathbb{Z}^d$ of infinite length.

1.4.1 Large Number Results

Let $\{X_n\}_{n \ge 1}$ be the simple symmetric random walk on $\mathbb{Z}^d$ of infinite length. Define

$$A = \{\omega \in \Omega \mid X_n(\omega)/n \to 0 \text{ as } n \to \infty\}. \tag{1.32}$$

Then, one can show that $\mathbb{P}(A) = 1$ despite $A \ne \Omega$. This is a consequence of *Stirling's law of large numbers*. Also,

$$\mathbb{P}\left(\frac{X_n}{\sqrt{n}} \le x\right) \to \int_{-\infty}^x \frac{e^{-y^2/2}}{\sqrt{2\pi}} dy \text{ as } n \to \infty. \tag{1.33}$$

for when $d = 1$. This is a consequence of the *central limit theorem*.

Chapter 2

MARKOV CHAINS

For the simple symmetric random walk on $\mathbb{Z}^d$, we saw that when $\mathbb{P}(X_1 = i_1, \ldots, X_n = i_n)$ is non-zero, it is given by

$$\mathbb{P}(X_1 = i_1, \ldots, X_n = i_n) = \mathbb{P}(X_1 = i_1) \prod_{k=2}^{n} \mathbb{P}(X_k = i_k \mid X_{k-1} = i_{k-1}) = \prod_{k=1}^{n} \mathbb{P}(X_2 = i_k \mid X_1 = i_{k-1}) \tag{2.1}$$

where $i_0 = 0$. If the walk were not symmetric, we could then define

$$p_{ij} = \begin{cases} p & \text{if } i = j + 1, \\ 1 - p & \text{if } i = j - 1, \\ 0 & \text{otherwise.} \end{cases} \tag{2.2}$$

With this, the above probability is $\prod_{k=1}^{n} p_{i_k i_{k-1}}$. In fact, the walk need not simple too, with steps of arbitrary size taking place. Moreover, $\mathbb{Z}^d$ can be replaced by any discrete space S. This is where we continue.

2.1 Discrete Time and Space Markov Chain

Let S be any countable set, called our state space, and X_n denote the state of the system at time n. The dynamics of the system is given by the transition probabilities; X_n moves from $i \in S$ to $j \in S$ with probability p_{ij}, which depends only on i and j. Some obvious properties include $p_{ij} \in [0, 1]$ and $\sum_{j \in S} p_{ij} = 1$. The transition probabilities can be arranged in a matrix $P = [p_{ij}]_{i,j \in S}$, called the one-step *transition matrix*.

Definition 2.1. Let μ be any probability on $(S, \mathcal{P}(S))$, and let $P = [p_{ij}]_{i,j \in S}$ be a one-step transition matrix. Moreover, let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with $(X_n)_{n \geq 0}$ a sequence of S-valued random variables whose joint distribution is given by

$$\mathbb{P}(X_0 = i_0, X_1 = i_1, \ldots, X_n = i_n) = \mu(\{i_0\}) \prod_{k=1}^{n} p_{i_{k-1} i_k}. \tag{2.3}$$

Then $\{X_n\}_{n \geq 0}$ is called a *Markov chain* on S with initial distribution μ and transition matrix P.

To find the distribution of X_n, we can use the law of total probability to get

$$\mathbb{P}(X_n = j) = \sum_{i_0, \ldots, i_{n-1} \in S} \mathbb{P}(X_0 = i_0, \ldots, X_{n-1} = i_{n-1}, X_n = j) = \sum_{i_0, \ldots, i_{n-1} \in S} \mu(\{i_0\}) \prod_{k=1}^{n} p_{i_{k-1} i_k}. \tag{2.4}$$

We can also show that the distribution of X_n when the previous $n-1$ steps are known, depends only on the last step; we have

$$\begin{aligned}\mathbb{P}(X_n = j \mid X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) &= \frac{\mathbb{P}(X_0 = i_0, \ldots, X_{n-1} = i_{n-1}, X_n = j)}{\mathbb{P}(X_0 = i_0, \ldots, X_{n-1} = i_{n-1})} \\ &= \frac{\mu(\{i_0\}) \prod_{k=1}^{n} p_{i_{k-1} i_k}}{\mu(\{i_0\}) \prod_{k=1}^{n-1} p_{i_{k-1} i_k}} = p_{i_{n-1} j}. \end{aligned} \tag{2.5}$$

On the other hand,

$$\begin{aligned}\mathbb{P}(X_n = j \mid X_{n-1} = i_{n-1}) &= \frac{\mathbb{P}(X_n = j, X_{n-1} = i_{n-1})}{\mathbb{P}(X_{n-1} = i_{n-1})} \\ &= \frac{\sum_{i_0, \ldots, i_{n-2} \in S} \mu(\{i_0\}) \prod_{k=1}^{n} p_{i_{k-1} i_k} p_{i_{n-1} j}}{\sum_{i_0, \ldots, i_{n-2} \in S} \mu(\{i_0\}) \prod_{k=1}^{n} p_{i_{k-1} i_k}} = p_{i_{n-1} j}. \end{aligned} \tag{2.6}$$

Thus, the distribution of X_n depends only on the previous state, and not on the entire history. Also, the distribution of X_n given the start state $X_0 = i_0$ is

$$\mathbb{P}(X_n = j \mid X_0 = i_0) = \sum_{i_1, \ldots, i_{n-1} \in S} \prod_{k=1}^{n} p_{i_{k-1} i_k}. \tag{2.7}$$

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If, in the distribution of X_n, the starting point $X_0 = i$ is specified, we can write the distribution as $\mathbb{P}(X_n = j \mid X_0 = i) = p_{ij}^n$, where p_{ij}^n is the (i,j)-th entry of the n-step transition matrix P^n. Further, if S is finite, then whenever $r + s = n$, $P^n = P^r P^s$, and thus $p_{ij}^n = \sum_{k \in S} p_{ik}^r p_{kj}^s$. Expanding gives

$$\mathbb{P}(X_n = j \mid X_0 = i) = \sum_{k \in S} \mathbb{P}(X_r = k \mid X_0 = i) \cdot \mathbb{P}(X_s = j \mid X_0 = k). \tag{2.8}$$

This is called the *Chapman-Kolmogorov equation*.

Example 2.2. Let S be a countable vertex set, and let $E \subseteq \{\{i,j\} \mid i,j \in S\}$ be a set of unordered edges. Let $\eta : E \to (0,\infty)$ be a function that assigns a positive weight to each edge. We define $\eta_{ij} = \eta(\{i,j\})$ for all $\{i,j\} \in E$, and $\eta_{ij} = 0$ if $\{i,j\} \notin E$. Then further define $\eta_i = \sum_{j \in S} \eta_{ij}$. We may assume that $\eta_i < \infty$ for all $i \in S$. Now, set $p_{ij} = \eta_{ij}/\eta_i$ for all $i,j \in S$. Then, $0 \le p_{ij} \le 1$ and $\sum_{j \in S} p_{ij} = 1$. Thus, $P = [p_{ij}]_{i,j \in S}$ is a transition matrix. The Markov chain with this transition matrix, and S as the state space, is simply a random walk on the weighted graph (S, E, η). In particular, if $\eta_{ij} = 1$ for all $\{i,j\} \in E$, then $p_{ij} = 1/\deg(i)$, where $\deg(i)$ is the degree of vertex i. This the simple random walk on the graph (S, E).

Example 2.3. Let $S = \{0,1\}$, and let $P = \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix}$ with $0 \le \alpha, \beta \le 1$. Also define the initial distribution as $\mu(\{0\}) = \mu_0$ and $\mu(\{1\}) = \mu_1 = 1 - \mu_0$. Then, $\{X_n\}_{n \ge 0}$ is a Markov chain on S with transition matrix P and initial distribution μ. This is called a two-state Markov chain.

Let us expand on the above example; the distribution of X_n is can be given as follows. We know $P^0 = I$ and $P^n = P^{n-1}P$ for $n \ge 1$. Thus, we can write

$$p_{00}^n = p_{00}^{n-1}(1-\alpha) + p_{01}^{n-1}\beta = p_{00}^{n-1}(1-\alpha-\beta) + \beta. \tag{2.9}$$

This is a linear recurrence relation; first checking for constant solutions $p_{00}^n \equiv p$ gives $p = \frac{\beta}{\alpha+\beta}$. For a more general solution, set $x_n = p_{00}^n - \frac{\beta}{\alpha+\beta}$, which gives $x_n = (1-\alpha-\beta)x_{n-1}$. Thus, $x_n = (1-\alpha-\beta)^n x_0$. With $x_0 = p_{00}^0 - \frac{\beta}{\alpha+\beta} = 1 - \frac{\beta}{\alpha+\beta} = \frac{\alpha}{\alpha+\beta}$, we have

$$p_{00}^n = (1-\alpha-\beta)^n \frac{\alpha}{\alpha+\beta} + \frac{\beta}{\alpha+\beta}. \tag{2.10}$$

One can also show that

$$p_{10}^n = \frac{\beta}{\alpha+\beta} - \frac{\beta}{\alpha+\beta}(1-\alpha-\beta)^n. \tag{2.11}$$

Thus, the distribution of X_n is given by

$$\begin{aligned}\mathbb{P}(X_n = 0) &= \mathbb{P}(X_n = 0 \mid X_0 = 0)\cdot\mathbb{P}(X_0 = 0) + \mathbb{P}(X_n = 0 \mid X_0 = 1)\cdot\mathbb{P}(X_0 = 1) = p_{00}^n\mu_0 + p_{10}^n\mu_1 \\ &= (p_{00}^n - p_{10}^n)\mu_0 + p_{10}^n = \frac{\beta}{\alpha+\beta} + (1-\alpha-\beta)^n\left(\mu_0 - \frac{\beta}{\alpha+\beta}\right),\end{aligned} \tag{2.12}$$

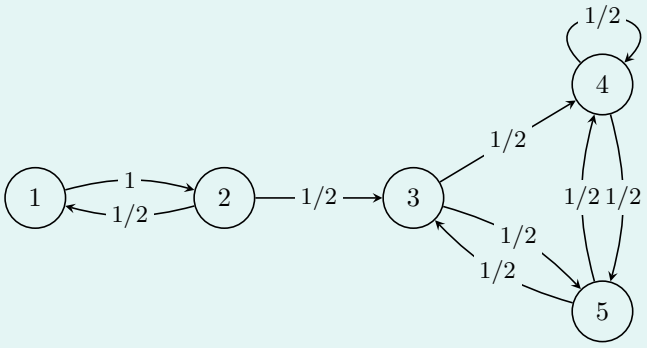
and

$$\mathbb{P}(X_n = 1) = 1 - \mathbb{P}(X_n = 0) = \frac{\alpha}{\alpha+\beta} - (1-\alpha-\beta)^n\left(\mu_1 - \frac{\alpha}{\alpha+\beta}\right). \tag{2.13}$$

Immediately, we can make some observations.

1. If we take $\mu_0 = \frac{\beta}{\alpha+\beta}$, then $\mathbb{P}(X_n = 0) = \frac{\beta}{\alpha+\beta}$ and $\mathbb{P}(X_n = 1) = \frac{\alpha}{\alpha+\beta}$ for all n. Thus, the distribution of X_n is independent of n.
2. If $\alpha+\beta > 0$, or $1-\alpha-\beta < 1$, then $\lim_{n\to\infty}(1-\alpha-\beta)^n = 0$. Thus, $\lim_{n\to\infty}\mathbb{P}(X_n = 0) = \frac{\beta}{\alpha+\beta}$ and $\lim_{n\to\infty}\mathbb{P}(X_n = 1) = \frac{\alpha}{\alpha+\beta}$.

Example 2.4. Consider the following weighted digraph.



Intuitively, it is clear that the Markov chain will spend a 'finite' amount of time in states 1 and 2, and an 'infinite' amount of time in states 3, 4, and 5. We can formalize this by computing the distribution of X_n as $n \to \infty$. We wish to show that $p_{11}^n \to 0$ as $n \to \infty$.

2.1.1 Recurrence and Transience

Let $\{X_n\}_{n\geq 0}$ be a Markov chain on a countable state space S with transition matrix $P = [p_{ij}]_{i,j\in S}$, and initial distribution μ. For all $n \geq 1$ and $i \in S$, define

$$N_n(i) = \sum_{k=0}^{n} \mathbf{1}_{\{X_k = i\}} \quad \text{and} \quad N(i) = \sum_{k=0}^{\infty} \mathbf{1}_{\{X_k = i\}}. \tag{2.14}$$

That is, $N_n(i)$ is the number of visits to state i up to time n, and $N(i)$ is the total number of visits to state i. From here, we can define the following.

Definition 2.5. A state $i \in S$ is called *recurrent* if $\mathbb{P}(N(i) = \infty) = 1$, and *transient* if $\mathbb{P}(N(i) < \infty) = 1$. The chain $\{X_n\}_{n\geq 1}$ is called recurrent if all states are recurrent, and transient if all states are transient.

We shall later show that every state in a Markov chain is either recurrent or transient. For $i \in S$, let $T_i = \min\{n \geq 1 \mid X_n = i\}$ be the first return time to state i.

Lemma 2.6. *Let $i \in S$, and $m \geq 1$. Then*

$$\mathbb{P}(N(i) \geq m \mid X_0 = i) = \mathbb{P}(T_i < \infty \mid X_0 = i)^m. \tag{2.15}$$

Proof. We have

$$\{N(i) \geq m\} = \bigcup_{k=1}^{\infty} \{N(i) \geq m,\, T_i = k\}. \tag{2.16}$$

Suppose, for $k \geq 1$, that $\mathbb{P}(T_i = k \mid X_0 = i) > 0$. Then, by the Markov property, we have

$$\begin{aligned}\mathbb{P}(N(i) \geq m,\, T_i = k \mid X_0 = i) &= \mathbb{P}(N(i) \geq m \mid T_i = k,\, X_0 = i) \cdot \mathbb{P}(T_i = k \mid X_0 = i) \\ &= \mathbb{P}(N(i) \geq m - 1 \mid X_0 = i) \cdot \mathbb{P}(T_i = k \mid X_0 = i).\end{aligned} \tag{2.17}$$

Note that this equation is trivially true if $\mathbb{P}(T_i = k \mid X_0 = i) = 0$. Thus, we have

$$\begin{aligned}\mathbb{P}(N(i) \geq m \mid X_0 = i) &= \sum_{k=1}^{\infty} \mathbb{P}(N(i) \geq m,\, T_i = k \mid X_0 = i) \\ &= \sum_{k=1}^{\infty} \mathbb{P}(N(i) \geq m - 1 \mid X_0 = i) \cdot \mathbb{P}(T_i = k \mid X_0 = i) \\ &= \mathbb{P}(N(i) \geq m - 1 \mid X_0 = i) \cdot \sum_{k=1}^{\infty} \mathbb{P}(T_i = k \mid X_0 = i) \\ &= \mathbb{P}(N(i) \geq m - 1 \mid X_0 = i) \cdot \mathbb{P}(T_i < \infty \mid X_0 = i).\end{aligned} \tag{2.18}$$

Inductively, we conclude that

$$\mathbb{P}(N(i) \geq m \mid X_0 = i) = \mathbb{P}(T_i < \infty \mid X_0 = i)^m. \tag{2.19}$$

■

Proposition 2.7. *Let $i \in S$. Then i is recurrent if and only if $\mathbb{P}(T_i < \infty \mid X_0 = i) = 1$, and transient if and only if $\mathbb{P}(T_i < \infty \mid X_0 = i) < 1$.*

Immediately from here, a state must be recurrent or transient.

Proof. Note that $\{N(i) = \infty\} = \bigcap_{m=1}^{\infty} \{N(i) \geq m\}$. Thus, by continuity of probability measures, we have

$$\mathbb{P}(N(i) = \infty) = \lim_{m \downarrow \infty} \mathbb{P}(N(i) \geq m). \tag{2.20}$$

Therefore,

$$\mathbb{P}(N(i) = \infty \mid X_0 = i) = \lim_{m \to \infty} \mathbb{P}(N(i) \geq m \mid X_0 = i) = \lim_{m \to \infty} \mathbb{P}(T_i < \infty \mid X_0 = i)^m \tag{2.21}$$

which is 1 if $\mathbb{P}(T_i < \infty \mid X_0 = i) = 1$, and 0 if $\mathbb{P}(T_i < \infty \mid X_0 = i) < 1$, by the previous lemma. Thus, i is recurrent if and only if $\mathbb{P}(T_i < \infty \mid X_0 = i) = 1$, and transient if and only if $\mathbb{P}(T_i < \infty \mid X_0 = i) < 1$. ■

Proposition 2.8. *Let $i \in S$. Then i is transient if and only if $\sum_{n=0}^{\infty} p_{ii}^n < \infty$, and recurrent if and only if $\sum_{n=0}^{\infty} p_{ii}^n = \infty$.*

Proof. Note that $\mathbb{P}(T_i < \infty \mid X_0 = i) < 1$ if and only if $\sum_{m=1}^{\infty} \mathbb{P}(T_i < \infty \mid X_0 = i)^m < \infty$, which is equivalent to $\sum_{m=1}^{\infty} \mathbb{P}(N(i) \geq m - 1 \mid X_0 = i) < \infty$. But, $\sum_{m=1}^{\infty} \mathbb{P}(N(i) \geq m - 1 \mid X_0 = i) = \sum_{m=0}^{\infty} \mathbb{P}(N(i) \geq m \mid X_0 = i) = \mathbb{E}[N(i) \mid X_0 = i]$. This last term can be rewritten as

$$\mathbb{E}[N(i) \mid X_0 = i] = \mathbb{E}\left[\sum_{n=1}^{\infty} \mathbf{1}_{\{X_n = i\}} \,\middle|\, X_0 = i\right] = \sum_{n=1}^{\infty} \mathbb{E}[\mathbf{1}_{\{X_n = i\}} \mid X_0 = i] = \sum_{n=1}^{\infty} p_{ii}^n. \tag{2.22}$$

■

Example 2.9. Consider the state space $S = \mathbb{Z}$ with $p_{ij} = 1/2$ if $j = i+1$ or $j = i-1$, and $p_{ij} = 0$ otherwise. This is the simple symmetric random walk on $\mathbb{Z}$. We claim that $0 \in S$ is recurrent. To see this, we have

$$p_{00}^{2n} = \frac{1}{2^{2n}}\binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}} \implies \sum_{n=0}^{\infty} p_{00}^{2n} = \infty. \tag{2.23}$$

Therefore, 0 is recurrent. Now, let $i \in \mathbb{Z}$ be any state. There exist $r, s \in \mathbb{N}$ such that $p_{i0}^r > 0$ and $p_{0i}^s > 0$. If we let $n \in \mathbb{N}$, then $n = r + s + k$ for some $k \geq 1$. Thus, $p_{ii}^n \geq p_{i0}^r p_{00}^k p_{0i}^s$. Therefore, $\sum_{n=0}^{\infty} p_{ii}^n \geq p_{i0}^r p_{0i}^s \sum_{k=0}^{\infty} p_{00}^k = \infty$, and thus i is recurrent. Since i was arbitrary, the chain is recurrent.

2.1.2 Hitting Probabilities

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Suppose we have two players, A and B, with an initial capital of two rupees. The dynamics are as follows: toss a fair coin. If heads, A wins and gets one rupee, and B loses one. If tails, B wins and gets one rupee, and A loses one. The game ends if either player runs out of money, and the other player is declared the winner. We can model this as a Markov chain with state space $S = \{0, 1, 2, 3, 4\}$, and $(X_n)_{n\geq 0}$ denoting the capital of A at time n with $X_0 = 2$. We wish to obtain the probability that A wins. To this end, define $T^4 = \min\{k \geq 0 \mid X_k = 4\}$. Thus, we need to compute the probability $\mathbb{P}(T^4 < \infty \mid X_0 = 2)$. If we define the map $h : S \to [0, 1]$ as $h(i) = \mathbb{P}(T^4 < \infty \mid X_0 = i)$, then we have $h(4) = 1$ and $h(0) = 0$. For $i \in \{1, 2, 3\}$, we have

$$\begin{aligned} h(i) &= \mathbb{P}(T^4 < \infty \mid X_0 = i) = \mathbb{P}(T^4 < \infty,\, X_1 = i+1 \mid X_0 = i) + \mathbb{P}(T^4 < \infty,\, X_1 = i-1 \mid X_0 = i) \\ &= \mathbb{P}(T^4 < \infty \mid X_1 = i+1,\, X_0 = i) \cdot \mathbb{P}(X_1 = i+1 \mid X_0 = i) \\ &\quad + \mathbb{P}(T^4 < \infty \mid X_1 = i-1,\, X_0 = i) \cdot \mathbb{P}(X_1 = i-1 \mid X_0 = i) \\ &= \frac{1}{2}\mathbb{P}(T^4 < \infty \mid X_1 = i+1) + \frac{1}{2}\mathbb{P}(T^4 < \infty \mid X_1 = i-1) = \frac{1}{2}h(i+1) + \frac{1}{2}h(i-1). \end{aligned} \tag{2.24}$$

Solving this recurrence relation with the boundary conditions $h(0) = 0$ and $h(4) = 1$, we get $h(i) = i/4$.

Now suppose A has an initial capital of 10 rupees, and instead plays against the casino. Moreover, they play with a biased coin, with the probability of heads being $p \in (0, 1)$, and $p \neq 1/2$. We question whether A is guaranteed to lose all their money. To this end, the state space is $S = \{0, 1, 2, \ldots\}$, and $(X_n)_{n\geq 0}$ denotes the capital of A at time n with $X_0 = 10$. We wish to compute the probability $\mathbb{P}(T^0 < \infty \mid X_0 = 10)$, where $T^0 = \min\{k \geq 0 \mid X_k = 0\}$. Define the map $h : S \to [0, 1]$ as $h(i) = \mathbb{P}(T^0 < \infty \mid X_0 = i)$. Then, we have $h(0) = 1$, and for $i \geq 1$, we have

$$h(i) = \mathbb{P}(T^0 < \infty \mid X_0 = i) = p \cdot h(i+1) + (1-p) \cdot h(i-1). \tag{2.25}$$

We can guess that $h(i) = \alpha^i$; this then leads to $h(i) = c_1 + c_2\left(\frac{1-p}{p}\right)^i$ for some $c_1, c_2 \geq 0$ for the recurrence relation. Using the boundary condition $h(0) = 1$, we have $c_1 + c_2 = 1$. Therefore,

$$h(i) = c_1 + (1 - c_1)\left(\frac{1-p}{p}\right)^i. \tag{2.26}$$

If $c_1 \neq 1$, and $p < 1/2$, then $\lim_{i\to\infty} h(i) = \infty$, which is impossible. Thus, $c_1 = 1$, and $h(i) = 1$ for all $i \geq 0$. Therefore, A is guaranteed to lose all their money. If $p > 1/2$, then $\lim_{i\to\infty} h(i) = c_1$; we will show later that this forces $c_1 = 0$, and thus $h(i) = \left(\frac{1-p}{p}\right)^i$ for all $i \geq 0$. Therefore, A is not guaranteed to lose all their money.

Theorem 2.10. *Let $(X_n)_{n\geq 0}$ be a Markov chain on S with transition matrix P and initial distribution μ. Let $A \subseteq S$, and define $T^A = \min\{n \geq 0 \mid X_n \in A\}$. Consider the map $h^A : S \to [0, 1]$ defined as*

$h^A(i) = \mathbb{P}(T^A < \infty \mid X_0 = i)$. *Then*

$$h^A(i) = \begin{cases} 1 & \text{if } i \in A, \\ \sum_{j \in S} p_{ij} h^A(j) & \text{if } i \notin A. \end{cases} \tag{2.27}$$

If $f : S \to [0, \infty)$ is any other map satisfying the above equation, then $h^A(i) \leq f(i)$ for all $i \in S$.

Example 2.11. Consider the simple symmetric random walk on $\mathbb{Z}^d$, with $P_{ij} = 1/(2d)$ if $|i - j| = 1$ and 0 otherwise. For $d = 1$, we showed that

$$p_{00}^{2n} = \frac{1}{2^{2n}} \binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}} \implies \sum_{n=0}^{\infty} p_{00}^{2n} = \infty, \tag{2.28}$$

and thus 0 (and every state in $\mathbb{Z}$) is recurrent. For $d = 2$, we have

$$p_{(0,0),(0,0)}^{2n} = \frac{1}{4^{2n}} \binom{2n}{n}^2 \sim \frac{1}{\pi n} \implies \sum_{n=0}^{\infty} p_{(0,0),(0,0)}^{2n} = \infty, \tag{2.29}$$

and thus 0 is recurent. Note that this example works easily since we are working in the $\mathbb{Z}^d$ lattice structure. If we, instead, replace $\mathbb{Z}^2$ with a hexagonal lattice, then using $\sum_{n \geq 0} p_{jj}^n = \infty$ is not as easy.

2.1.3 Weighted Graphs

Recall **Example 2.2**, where we had a weighted graph $\Gamma = (G, \mu)$, with $G = (V, E)$ and $\mu : E \to (0, \infty)$. We defined a Markov chain on V with transition matrix $P = [p_{ij}]_{i,j \in S}$, where $p_{ij} = \mu_{ij}/\mu_i$ for all $i, j \in V$. Here, notice that $p_{ij} = p_{ji}$ for all $i, j \in V$. Also, for $x_0, x_1, \ldots, x_n \in V$, we have

$$\mu_{x_0} \mathbb{P}(X_0 = x_0, \ldots, X_n = x_n \mid X_0 = x_0) = \prod_{i=1}^{n} \mu_{x_{i-1}, x_i} = \mu_{x_n} \mathbb{P}(X_0 = x_n, \ldots, X_n = x_0 \mid X_0 = x_n). \tag{2.30}$$

This is known as the *time reversible property*.

Theorem 2.12. *Let Γ be a weighted graph with vertex set V and weight function $\mu : E \to (0, \infty)$. Also let $(X_n)_{n \geq 0}$ be the random walk on Γ. Then $\sum_{i \in V} \mu_i < \infty$ implies that $(X_n)_{n \geq 0}$ is recurrent.*

Proof. Let $x \in V$. Let $h : V \to [0, 1]$ be defined as $h(i) = \mathbb{P}(T^x < \infty \mid X_0 = i)$, where $T^x = \min\{n \geq 0 \mid X_n = x\}$. We shall show that $h \equiv 1$. We know that $h(x) = 1$, and for $i \neq x$, we have $h(i) = \sum_{j \in V} p_{ij} h(j)$. We work with the energy principle. For $\varphi : V \to [0, \infty)$, define

$$\mathcal{E}(\varphi, \varphi) = \frac{1}{2} \sum_{z,w \in V} \mu_{zw} (\varphi(z) - \varphi(w))^2. \tag{2.31}$$

If we set $\varphi = 1 - h$, then showing $\mathcal{E}(\varphi, \varphi) = 0$ gives $h \equiv 1$. Now

$$\mathcal{E}(\varphi, \varphi) = \frac{1}{2} \sum_{z,w \in V} \mu_{zw} (\varphi(z) - \varphi(w))^2 \tag{2.32}$$

$$= \sum_{z \in V} \varphi(z) \sum_{w \in V} \mu_{zw} (\varphi(z) - \varphi(w)). \tag{2.33}$$

For $z \neq x$, we have $\varphi(z) = \sum_{w \in V} p_{zw} \varphi(w)$, so

$$\sum_{w \in V} \mu_{zw} (\varphi(z) - \varphi(w)) = \mu_z \left(\varphi(z) - \sum_{w \in V} p_{zw} \varphi(w) \right) = 0. \tag{2.34}$$

Also $\varphi(x) = 1 - h(x) = 0$, so the $z = x$ term vanishes as well. Hence $\mathcal{E}(\varphi, \varphi) = 0$. Since each summand is nonnegative, $\varphi(z) = \varphi(w)$ on every edge with $\mu_{zw} > 0$; connectedness then implies φ is constant on V. Because $\varphi(x) = 0$, that constant is 0. Thus $h(i) = 1$ for all $i \in V$, and therefore

$$\mathbb{P}(T^x < \infty \mid X_0 = i) = 1 \qquad \text{for all } i \in V. \tag{2.35}$$

So the walk hits x almost surely from any starting point, and hence x is recurrent. Since x was arbitrary, the random walk is recurrent. ■

For a weghted graph $\Gamma = (G, \mu)$, on a finite set of vertices, denote $j \leftarrow i$ if there exists a natural n such that $p_{ij}^n > 0$. That is, j is reachable from i. Further, denote $j \leftrightarrow i$ if $j \leftarrow i$ and $i \leftarrow j$. Here, we say that i and j communicate. Now define

$$T = \{j \in S \mid \text{ there exists } k \in S \text{ such that } k \leftarrow j \text{ but } j \not\leftarrow k\}. \tag{2.36}$$

That is, all states in T can reach some state, but cannot be reached from that state. If $S = T$, then this happens to be a contradiction to the fact that S is finite. Thus, $T \subsetneq S$. Let $z_1 \in S \setminus T$, and define $R_1 = \{x \in S \mid z_1 \leftrightarrow x\}$. If $S = T \sqcup R_1$, then we are done. Otherwise, let $z_2 \in S \setminus (T \sqcup R_1)$, and define $R_2 = \{x \in S \mid z_2 \leftrightarrow x\}$. Continuing in this manner, we can write

$$S = T \sqcup R_1 \sqcup R_2 \sqcup \cdots \sqcup R_k, \quad R_i = \{x \in S \mid z_i \leftrightarrow x\}, \tag{2.37}$$

for some $k \geq 1$. Note that $R_i \cap R_j = \emptyset$ for $i \neq j$, and $R_i \cap T = \emptyset$ for all i. Take $y \in R_i$ for some $i = 1, 2, \ldots, k$. Let $n_0(y) \in \mathbb{N}$ be such that $p_{y,z_i}^{n_0} > 0$. Also let $c = \min_{y \in R_i} p_{y,z_i}^{n_0(y)} > 0$. Then,

$$\begin{aligned}
\sum_{y \in R_i} \sum_{m=0}^{\infty} p_{yy}^m &\geq \sum_{y \in R_i} \sum_{m=n_0(y)+1}^{\infty} p_{yy}^m \geq \sum_{y \in R_i} \sum_{m=n_0(y)+1}^{\infty} p_{y,z_i}^{n_0(y)} p_{z_i,y}^{m-n_0(y)} \\
&= \sum_{y \in R_i} p_{y,z_i}^{n_0(y)} \left(\sum_{k=1}^{\infty} p_{z_i,y}^k \right) \geq c \sum_{y \in R_i} \sum_{k=1}^{\infty} p_{z_i,y}^k = c \sum_{k=1}^{\infty} \sum_{y \in R_i} p_{z_i,y}^k = c \sum_{k=1}^{\infty} 1 = \infty.
\end{aligned} \tag{2.38}$$

Since $|R_i| < \infty$, there exists $y \in R_i$ such that $\sum_{m=0}^{\infty} p_{yy}^m = \infty$. Thus, y is recurrent. But any other $y \in R_i$ communicates with y, and thus is also recurrent. Therefore, all states in R_i are recurrent. Since i was arbitrary, we conclude that all states in $\bigcup_{i=1}^k R_i$ are recurrent. Now let $i \in T$. By definition, there exists $j \in S$ such that $p_{ji}^k = 0$ for all $k \geq 1$. Thus, $\mathbb{P}(T^i < \infty \mid X_0 = j) = 0$, and thus i is transient. Therefore, we have shown that all states in S are either recurrent or transient.

September 2nd.

Proposition 2.13. *Let $(X_n)_{n \geq 1}$ be a Markov chain with with transition matrix P on state space S.*

1. *Let π be a stationary distribution, that is, $\pi P = \pi$ and $\sum_{i \in S} \pi_i = 1$. Then, if $X_0 \overset{d}{=} \pi$, we have $X_n \overset{d}{=} \pi$ for all $n \geq 1$.*
2. *If S is finite, then any limiting distribution must be a stationary distribution.*
3. *If S is finite, and P is doubly stochastic, that is, $\sum_{i \in S} p_{ij} = 1$ for all $j \in S$, then the uniform distribution on S is the stationary distribution.*

Proof. 1. We show via induction. For $n = 1$, we have

$$\mathbb{P}(X_1 = j) = \sum_{i \in S} \mathbb{P}(X_1 = j \mid X_0 = i) \cdot \mathbb{P}(X_0 = i) = \sum_{i \in S} p_{ij} \pi(i) = \pi_j \tag{2.39}$$

since $\pi P = \pi$. Now suppose that $X_n \overset{d}{=} \pi$. Then, we have

$$\mathbb{P}(X_{n+1} = j) = \sum_{i \in S} \mathbb{P}(X_{n+1} = j \mid X_n = i) \cdot \mathbb{P}(X_n = i) = \sum_{i \in S} p_{ij} \pi(i) = \pi_j. \tag{2.40}$$

2. Fix $i \in S$. Define $\pi(j) := \lim_{n\to\infty} p_{ij}^n$ for all $j \in S$. Then $0 \le \pi(j) \le 1$ for all $j \in S$, and

$$\sum_{j\in S} \pi(j) = \sum_{j\in S} \lim_{n\to\infty} p_{ij}^n = \lim_{n\to\infty} \sum_{j\in S} p_{ij}^n = 1. \tag{2.41}$$

For $n \ge 2$, we know that $p_{ij}^n = \sum_{k\in S} p_{ik}^{n-1} p_{kj}$. Thus, taking limits, we have

$$\pi(j) = \lim_{n\to\infty} p_{ij}^n = \lim_{n\to\infty} \sum_{k\in S} p_{ik}^{n-1} p_{kj} = \sum_{k\in S} \lim_{n\to\infty} p_{ik}^{n-1} p_{kj} = \sum_{k\in S} \pi(k) p_{kj} \tag{2.42}$$

or, equivalently, $\pi P = \pi$. Therefore, π is a stationary distribution.

3. Let π be the uniform distribution on S; that is, $\pi(i) = 1/|S|$ for all $i \in S$. Then, we have

$$\sum_{k\in S} \pi(k) p_{kj} = \frac{1}{|S|} \sum_{k\in S} p_{kj} = \frac{1}{|S|} = \pi(j). \tag{2.43}$$

π is a stationary distribution.

■

Theorem 2.14. *Let $(X_n)_{n\ge 0}$ be an irreducible Markov chain; that is, for all $i, j \in S$, there exists $n \ge 1$ such that $p_{ij}^n > 0$. Also let the state space S be finite. Then there exists a unique stationary distribution π on S.*

Proof. We utilize graph theory. Let $i \in S$. Construct a directed rooted tree (g) with a root in i, and with vertex set S, such that the following holds: for all $j \in S$, there exists a unique directed path from j to i. The root i has no outgoing edges, and all other vertices have exactly one outgoing edge. Let $G(i)$ denote the set of all such directed rooted trees with root in i. For all $i \in S$, define

$$\pi_0(i) := \sum_{g\in G(i)} \prod_{(j\to k)\in g} p_{jk}. \tag{2.44}$$

Then, we have

$$\pi_0(i) \left(\sum_{j\ne i} p_{ij} \right) = \sum_{j\ne i} \sum_{g\in G(i)} p_{ij} \prod_{(k\to l)\in g} p_{kl}. \tag{2.45}$$

We claim that this is equal to $\sum_{g\in L} \prod_{(j\to k)\in g} p_{jk}$ where L:

1. for all $k \in S$, there is only one directed edge pointing away from k to l, for $k \ne l \in S$, and
2. the graph has exactly one closed cycle with i in the cycle.

Set $C = \sum_{i\in S} \pi_0(i)$, and $\tilde{\pi}(i) = \pi_0(i)/C$.

For uniqueness, let π_1 and π_2 be two stationary distributions, and set $\pi = \pi_1 - \pi_2$. Then, we have $\pi P = \pi$ or $(P^\top - I)\pi^\top = 0$; $\pi^\top$ is in the null space of $P^\top - I$. Let $h : S \to \mathbb{R}$ be such that $Ph = h$. Also, let $M = \max\{h(i) \mid i \in S\}$, so that $h(i_0) = M$ for some $i_0 \in S$. Suppose $p_{i_0 j} > 0$ for some $j \in S$ with $h(j) < M$. Then, we have

$$h(i_0) = \sum_{k\in S} p_{ki_0} h(k) = h(j) p_{ji_0} + \sum_{k\ne j} p_{ki_0} h(k) < M p_{ji_0} + M \sum_{k\ne j} p_{ki_0} = M, \tag{2.46}$$

which is a contradiction. Thus, since the chain is irreducible, we have $h(j) = M$ for all $j \in S$. Therefore, the null space of $P^\top - I$ looks like $\{c \cdot \mathbf{1} \mid c \in \mathbb{R}\}$, where $\mathbf{1}$ is the vector of all ones. Thus, $\pi^\top$ is a scalar multiple of $\mathbf{1}$; this, along with that $\sum_{i\in S} \pi(i) = 0$, gives $\pi \equiv 0$. Therefore, $\pi_1 = \pi_2$, and the stationary distribution is unique. ■

Theorem 2.15. *Let $(X_n)_{n\ge 0}$ be an irreducible Markov chain on a finite state space S, and suppose*

that $X_0 \overset{d}{=} \pi$, where π is the unique stationary distribution. Then

$$\sup_{i,j\in S} \left|p_{ij}^n - \pi(j)\right| \leq C\Theta^n \tag{2.47}$$

for some $C > 0$ and $\Theta \in (0,1)$.

Proof. By previous theorem, we know that there exists a unique stationary distribution π on S. Note that, for $i \in S$, there exists $n_i \geq 1$ such that $p_{ii}^{n_i} > 0$. Fix such an i, and let $i \neq j \in S$. By irreducibility, there exists $m_j \geq 1$ such that $p_{ij}^{m_j} > 0$. Let $r = \max\{n_i \mid i \in S\} + \max\{m_j \mid j \in S\}$. Then $p_{ij}^r \geq p_{ii}^{n_i} p_{ij}^{m_j} > 0$. That is, we have obtained a $r \geq 1$ such that $p_{ij}^r > 0$ for all $i, j \in S$. Define $\Pi = [\pi(j)]_{i,j\in S}$, and let $1 > \delta > 0$. Then $p_{ij}^r \geq \delta\pi(i) > 0$ for all $i, j \in S$. With $\Theta = 1 - \delta$, define $Q = \frac{1}{\Theta}(P^r + (1-\Theta)\Pi)$. Then Q is a stochastic matrix, and $Q^k\Pi = \Pi$ for all $k \geq 1$. Since $\Pi P^r = \Pi$ also holds, we can rearrange to get $P^{rk} = \Theta^k Q^k + (1-\Theta^k)\Pi$. Now, for all $n \geq r$, there exist $k \geq 1$ and $0 \leq l < r$ such that $n = rk + l$. Then, we have

$$P^n = P^{rk+l} = \Theta^k Q^k P^l + (1-\Theta^k)\Pi P^l = \Theta^k Q^k P^l + (1-\Theta^k)\Pi \implies P^{rk+l} - \Pi = \Theta^k(Q^k P^l - \Pi). \tag{2.48}$$

Looking at the ij-th entry, we have

$$p_{ij}^{rk+l} - \pi(j) = \Theta^k (Q^k P^l)_{ij} - \pi(j) \implies \left|p_{ij}^n - \pi(j)\right| \leq C\Theta^k. \tag{2.49}$$

■

Theorem 2.16. *Let $(X_n)_{n\geq 0}$ be an irreducible aperiodic Markov chain on S, and assume there exists a stationary distribution π on S. Then, for all $i, j \in S$, we have*

$$\lim_{n\to\infty} p_{ij}^n = \pi(j). \tag{2.50}$$

Definition 2.17. A probability distribution π on S is called reversible for a Markov chain with transition matrix P if $\pi(i)p_{ij} = \pi(j)p_{ji}$ for all $i, j \in S$.

Suppose π is reversible for a Markov chain with transition matrix P. Then, we have

$$\sum_{j\in S} \pi(j)p_{ji} = \sum_{j\in S} \pi(i)p_{ij} = \pi(i)\sum_{j\in S} p_{ij} = \pi(i). \tag{2.51}$$

That is, π is a stationary distribution. Therefore, reversibility implies stationarity. However, the converse is not true in general.

Chapter 3

MARTINGALE THEORY

3.1 Discrete Time Martingales

September 16th.

Definition 3.1. A sequence of random variables $\{Z_n\}_{n\geq 1}$ is a *martingale* if

1. $\mathbb{E}[|Z_n|] < \infty$ for all $n \geq 1$,
2. $\mathbb{E}[Z_n \mid Z_{n-1} = z_{n-1}, \ldots, Z_1 = z_1] = z_{n-1}$.

To understand the second condition, let us focus on the following: with $(\Omega, \mathcal{F}, \mathbb{P})$ as a probability space, and $Y, X : \Omega \to S$ be discrete random variables with S a countable set, then with $y \in Y(\Omega)$, we have

$$\mathbb{E}[X \mid Y = y] = \sum_{k\in S} k\mathbb{P}(X = k \mid Y = y) \tag{3.1}$$

For multiple discrete random variables $Y_1, Y_2, \ldots, Y_n$, we have

$$\mathbb{E}[X \mid Y_1 = y_1, \ldots, Y_n = y_n] = \sum_{k\in S} k\mathbb{P}(X = k \mid Y_1 = y_1, \ldots, Y_n = y_n) \tag{3.2}$$

for $y_i \in Y_i(\Omega)$.

Example 3.2. Let $\{S_n\}_{n\geq 1}$ be the simple symmetric random walk on $\mathbb{Z}$, with $S_0 = 0$, and let $s_1, \ldots, s_{n-1} \in Z$ be such that $\mathbb{P}(S_{n-1} = s_{n-1}, \ldots, S_1 = s_1) > 0$. As a formal or imprecise calculation, we have

$$\begin{aligned}
\mathbb{E}[S_n \mid S_{n-1} = s_{n-1}, \ldots, S_1 = s_1] &= \sum_{k\in\mathbb{Z}} k\mathbb{P}(S_n = k \mid S_{n-1} = s_{n-1}, \ldots, S_1 = s_1) \\
&= \sum_{k=-n}^{n} k\mathbb{P}(S_{n-1} + X_n = k \mid S_{n-1} = s_{n-1}, \ldots, S_1 = s_1) \\
&= \sum_{k=-n}^{n} k\mathbb{P}(X_n = k - s_{n-1} \mid S_{n-1} = s_{n-1}, \ldots, S_1 = s_1) \\
&= \sum_{k=-n}^{n} k\mathbb{P}(X_n = k - s_{n-1}) \\
&= (s_{n-1} - 1)\mathbb{P}(X_n = 1) + (s_{n-1} + 1)\mathbb{P}(X_n = -1) = s_{n-1}. \qquad (3.3)
\end{aligned}$$

Also, $\mathbb{E}[|S_n|] < \infty$ for all $n \geq 1$. If instead, we consider the simple asymmetric random walk on $\mathbb{Z}$

(that is, $\mathbb{P}(X_n = 1) = p = 1 - \mathbb{P}(X_n = -1)$ and $p \neq \frac{1}{2}$), then we have

$$\mathbb{E}[S_n \mid S_{n-1} = s_{n-1}, \ldots, S_1 = s_1] = (s_{n-1} - 1)p + (s_{n-1} + 1)(1 - p) = s_{n-1} + 1 - 2p \neq s_{n-1}. \quad (3.4)$$

Hence, the simple symmetric random walk is a martingale, while the simple asymmetric random walk is not.

Example 3.3. Let X_n be a sequence of independent and identically distributed random variables with $\mathbb{P}(X_n = 2) = \frac{1}{2} = \mathbb{P}(X_n = 0)$, and then define $Z_n = \prod_{i=1}^{n} X_i$. Then, we have

$$\mathbb{E}[|Z_n|] = \prod_{k=1}^{n} \mathbb{E}[|X_k|] = \prod_{k=1}^{n} 1 = 1 < \infty \quad (3.5)$$

for all $n \geq 1$. Now, for $z_1, \ldots, z_{n-1} \in \{0, 2, 4, \ldots, 2^{n-1}\}$ such that $\mathbb{P}(Z_{n-1} = z_{n-1}, \ldots, Z_1 = z_1) > 0$, we have

$$\begin{aligned} \mathbb{E}[Z_n \mid Z_{n-1} = z_{n-1}, \ldots, Z_1 = z_1] &= \mathbb{E}[X_n Z_{n-1} \mid Z_{n-1} = z_{n-1}, \ldots, Z_1 = z_1] \\ &= z_{n-1}\mathbb{E}[X_n \mid Z_{n-1} = z_{n-1}, \ldots, Z_1 = z_1] \\ &= z_{n-1}\mathbb{E}[X_n] = z_{n-1}. \end{aligned} \quad (3.6)$$

Hence, $\{Z_n\}_{n\geq 1}$ is a martingale.

Suppose $\{Z_n\}_{n\geq 1}$ is a Markov chain. A consequence of the Markov property,

$$\mathbb{P}(Z_n = z_n, \ldots, Z_0 = z_0) = \mu(\{z_0\}) \prod_{k=1}^{n} p_{z_{k-1}, z_k}, \quad (3.7)$$

is that the past and the future are independent given the present. A martingale is less rigid than a Markov chain, saying that the expectation of the future given the present and the past is equal to the present. Moreover, Martingales are easier to construct and verify than Markov chains, and several limit results follow too.

Example 3.4. Let Z_1 be the random variable defined by $\mathbb{P}(Z_1 = 1) = \frac{1}{2} = \mathbb{P}(Z_1 = 0)$. If $\{X_n\}_{n\geq 1}$ is a sequence of independent and identically distributed random variables with $\mathbb{P}(X_n = 1) = \frac{1}{2} = \mathbb{P}(X_n = -1)$, then define $Z_n = Z_{n-1} + X_n Z_1$ for $n \geq 2$. Then, verify that $\{Z_n\}_{n\geq 1}$ is a Markov chain, but not a martingale.

From here on, we will interpet $\mathbb{E}[Z_n \mid z_{n-1}, \ldots, z_1]$ to be $\mathbb{E}[Z_n \mid Z_{n-1} = z_{n-1}, \ldots, Z_1 = z_1]$.

Let $\{Z_n\}_{n\geq 1}$ be a collection of discrete random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. Then we know that

$$\mathbb{E}[Z_n \mid Z_{n-1} = z_{n-1}, \ldots, Z_1 = z_1] = \sum_{k \in Z_n(\Omega)} k\mathbb{P}(Z_n = k \mid Z_{n-1} = z_{n-1}, \ldots, Z_1 = z_1). \quad (3.8)$$

Note that both sides of the above equation is a function of $z_1, \ldots, z_{n-1}$, say, $f(z_1, \ldots, z_{n-1})$. If $\{Z_n\}_{n\geq 1}$ were a martingale, then the equation above implies that $f(z_1, \ldots, z_{n-1}) = z_{n-1}$, which is a function of z_{n-1} alone. Thus, we define $Y_{n-1} := f(z_1, \ldots, z_{n-1})$, which we interpret as $\mathbb{E}[Z_n \mid z_{n-1}, \ldots, z_1]$.

Remark 3.5. We recall two important facts.

1. If Y and Z are discrete random variables, and $X = h(Y, Z)$ for some function $h : \mathbb{R}^2 \to \mathbb{R}$, then $\mathbb{E}[X \mid Y = y, Z = z] = h(y, z)$.
2. If X and Y are independent discrete random variables, then $\mathbb{E}[X \mid Y = y] = \mathbb{E}[X] \equiv \mathbb{E}[X \mid Y]$.

Example 3.6. Let $\{X_n\}_{n\geq 1}$ be a sequence of independent and identically distributed random variables with $\mathbb{P}(X_n = 1) = \frac{1}{2} = \mathbb{P}(X_n = -1)$. Define $S_n = \sum_{i=1}^{n} X_i$ for $n \geq 1$, and $S_0 = 0$. Now,

define $\tilde{Z}_n = e^{rS_n}$ for all $n \geq 1$, where $r \in \mathbb{R}$ is a constant. Then, we have

$$\mathbb{E}[\left|\tilde{Z}_n\right|] = \mathbb{E}[e^{rS_n}] \leq c < \infty \tag{3.9}$$

as $|S_n| \leq n$. Now, notice that $e^{rS_n} = e^{r\sum_{i=1}^n X_i} = e^{r\sum_{i=1}^n X_i}$. But with $\tilde{Y}_i = e^{rX_i}$, we have $\tilde{Y}_1, \ldots, \tilde{Y}_n$ are independent and identically distributed random variables, and $\tilde{Z}_n = \prod_{i=1}^n \tilde{Y}_i$. This gives $\mathbb{E}[X_i] = \frac{e^r + e^{-r}}{2} \neq 1$ unless $r = 0$. Hence, $\tilde{Z}_n$ is not a martingale. To extract a martingale out of this, define

$$q(r) = \log \mathbb{E}[\tilde{Y}_i] = \log\left(\frac{e^r + e^{-r}}{2}\right), \tag{3.10}$$

and let $Y_i = e^{rX_i - q(r)}$. Then, we have that Y_i's are independent and identically distributed random variables with $\mathbb{E}[Y_i] = 1$. Now, define $Z_n = \prod_{i=1}^n Y_i = e^{rS_n - nq(r)}$. Then, we have, via the same argument as in **Example 3.3**, that $\{Z_n\}_{n\geq 1}$ is a (product) martingale. More generally, if $\{X_n\}_{n\geq 1}$ is a sequence of independent and identically distributed random variables, and $S_n = \sum_{i=1}^n X_i$, then $\{Z_n\}_{n\geq 1}$ defined by $Z_n = e^{rS_n - nq(r)}$ is a martingale, where $q(r) = \log \mathbb{E}[e^{rX_1}]$.

Lemma 3.7. *Let $\{Z_n\}_{n\geq 1}$ be a martingale. Then, for any $n \geq 1$, we have*

$$\mathbb{E}[Z_n \mid z_i, z_{i-1}, \ldots, z_1] = z_i \tag{3.11}$$

for all $1 \leq i \leq n$.

Proof. We make use of the tower property of conditional expectation, which states that for any random variables X, Y, Z, we have

$$\mathbb{E}[\mathbb{E}[X \mid Y, Z] \mid Y] = \mathbb{E}[X \mid Y]. \tag{3.12}$$

We prove via induction. The base case is $n = 1$, which is trivial. Now, assume the statement holds for $n = m$, and let $n = m + 1$. Then,

$$\begin{aligned}\mathbb{E}[Z_{m+1} \mid z_i, z_{i-1}, \ldots, z_1] &= \mathbb{E}[\mathbb{E}[Z_{m+1} \mid z_m, z_{m-1}, \ldots, z_1] \mid z_i, z_{i-1}, \ldots, z_1] \\ &= \mathbb{E}[Z_m \mid z_i, z_{i-1}, \ldots, z_1] = z_i,\end{aligned} \tag{3.13}$$

where the second equality follows from the martingale property, and the third equality follows from the induction hypothesis. Hence, by induction, the statement holds for all $n \geq 1$. ■

If we apply the above lemma to $i = 1$, then

$$\mathbb{E}[Z_n \mid Z_1] = Z_1 \implies \mathbb{E}[Z_n] = \mathbb{E}[\mathbb{E}[Z_n \mid Z_1]] = \mathbb{E}[Z_1], \tag{3.14}$$

for all $n \geq 1$. Hence, the expectation of a martingale is constant over time.

Remark 3.8. We show the tower property of conditional expectation for discrete random variables. Let X, Y, Z be discrete random variables, and let $y \in Y(\Omega)$ and $z \in Z(\Omega)$. Let

$$h(y, z) = \mathbb{E}[X \mid Y = y, Z = z] = \sum_{k \in X(\Omega)} k\mathbb{P}(X = k \mid Y = y, Z = z). \tag{3.15}$$

Then,

$$\begin{aligned}\mathbb{E}[h(Y,Z)\mid Y=y_0] &= \sum_{y\in Y(\Omega)}\sum_{z\in Z(\Omega)} h(y,z)\mathbb{P}(Y=y,Z=z\mid Y=y_0) = \sum_{z\in Z(\Omega)} h(y_0,z)\mathbb{P}(Z=z\mid Y=y_0)\\ &= \sum_{z\in Z(\Omega)}\left(\sum_{k\in X(\Omega)} k\mathbb{P}(X=k\mid Y=y_0,Z=z)\right)\mathbb{P}(Z=z\mid Y=y_0)\\ &= \sum_{z\in Z(\Omega)}\sum_{k\in X(\Omega)} k\cdot\frac{\mathbb{P}(X=k,Y=y_0,Z=z)}{\mathbb{P}(Y=y_0)}\\ &= \frac{1}{\mathbb{P}(Y=y_0)}\sum_{z\in Z(\Omega)}\sum_{k\in X(\Omega)} k\mathbb{P}(X=k,Y=y_0,Z=z)\\ &= \frac{1}{\mathbb{P}(Y=y_0)}\sum_{k\in X(\Omega)} k\mathbb{P}(X=k,Y=y_0) = \mathbb{E}[X\mid Y=y_0]. \end{aligned} \tag{3.16}$$

Hence, $\mathbb{E}[h(Y,Z)\mid Y=y_0]=\mathbb{E}[X\mid Y=y_0]$, which is the tower property of conditional expectation.

September 30th.

Example 3.9. Suppose $\{X_i^{(n)}\}_{i,n\geq 1}$ is a collection of independent and identically distributed random variables with $\mathbb{P}(X_i^{(n)}=0)=\frac{1}{2}=\mathbb{P}(X_i^{(n)}=2)$, and define $Z_1=1$ and $Z_n=\sum_{i=1}^{Z_{n-1}} X_i^{(n)}$ for $n\geq 2$. Each particle in generation $n-1$ produces a random number of particles in generation n, and the number of particles produced by each particle is independent and identically distributed. We have $\mathbb{E}[|Z_1|]=1<\infty$, and

$$\mathbb{E}[|Z_{k+1}|]=\mathbb{E}\left[\left|\sum_{i=1}^{Z_k} X_i^{(k+1)}\right|\right]\leq \mathbb{E}\left[\sum_{i=1}^{Z_k}\left|X_i^{(k+1)}\right|\right]\leq 2\mathbb{E}[Z_k]<\infty \tag{3.17}$$

via induction. Now, assume $\mathbb{P}(Z_{n-1}=z_{n-1},\ldots,Z_1=z_1)>0$. Then

$$\mathbb{E}[Z_n\mid z_{n-1},\ldots,z_1]=\mathbb{E}\left[\sum_{i=1}^{Z_{n-1}} X_i^{(n)}\mid z_{n-1},\ldots,z_1\right]=\mathbb{E}\left[\sum_{i=1}^{z_{n-1}} X_i^{(n)}\right]=z_{n-1}\mathbb{E}[X_1^{(n)}]=z_{n-1}. \tag{3.18}$$

Therefore, $\{Z_n\}_{n\geq 1}$ is a martingale. This is an example of a branching process.

3.2 Stopped Processes

Let $X:\Omega\to\mathbb{R}$ be a discrete random variable over a probability space $(\Omega,\mathcal{F},\mathbb{P})$, and let $\mathcal{A}_X=\{X^{-1}(B)\mid B\in\mathcal{B}(\mathbb{R})\}$ be the set of all events observable by X; the set of all events E such that it may be written as the union of basic events $\{X=c\}$ with c coming from $X(\Omega)$.

Suppose $Z_1,\ldots,Z_n$ are discrete random variables. Let $\mathcal{A}_n$ be the set of all events observable by $Z_1,\ldots,Z_n$, or observable by time k; that is, the set of all events $E\subseteq\Omega$ such that it may be written as the union of basic events $\{Z_1=z_1,\ldots,Z_{n-1}=z_{n-1}\}$ where $z_i\in Z_i(\Omega)$.

We will adopt the above notation.

Definition 3.10. Let $\{Z_n\}_{n\geq 1}$ be a sequence of random variables. A map $T:\Omega\to\mathbb{N}\cup\{\infty\}$ is termed a *stopping time* if $\{T=k\}\in\mathcal{A}_k$, where $\mathcal{A}_k$ is the set of all event observable by time k.

Definition 3.11. For any stopping time T for a sequence $\{Z_n\}_{n\geq 1}$ of random variables, the *stopped*

process Z_n^T is defined as

$$Z_n^T = Z_{n\wedge T} \begin{cases} Z_n & \text{if } n < T, \\ Z_T & \text{if } n \geq T. \end{cases} \tag{3.19}$$

Theorem 3.12. *Given a sequence* $\{Z_n\}_{n\geq 1}$ *of random variables and a stopping time* T*, the stopped process* $\{Z_n^T\}_{n\geq 1}$ *is a martingale whenever* $\{Z_n\}_{n\geq 1}$ *is a martingale.*

Proof. Note that we may write $Z_n^T = \sum_{i=1}^{n-1} Z_i \mathbf{1}_{\{T=i\}} + Z_n \mathbf{1}_{\{T\geq n\}}$. Then, we have

$$\mathbb{E}[|Z_n^T|] \leq \sum_{k=1}^{n} \mathbb{E}[|Z_k|] < \infty \tag{3.20}$$

as $\{Z_n\}_{n\geq 1}$ is a martingale. Moreover,

$$\begin{aligned} \mathbb{E}[Z_n^T \mid \mathcal{A}_{n-1}] &= \mathbb{E}[\sum_{i=1}^{n-1} Z_i \mathbf{1}_{\{T=i\}} \mid \mathcal{A}_{n-1}] + \mathbb{E}[Z_n \mathbf{1}_{T\geq n} \mid \mathcal{A}_{n-1}] = \sum_{i=1}^{n-1} Z_i \mathbf{1}_{\{T=i\}} + \mathbf{1}_{T\geq n}\mathbb{E}[Z_n \mid \mathcal{A}_{n-1}] \\ &= \sum_{i=1}^{n-1} Z_i \mathbf{1}_{\{T=i\}} + \mathbf{1}_{T\geq n} Z_{n-1} = \sum_{i=1}^{n-2} Z_i \mathbf{1}_{\{T=i\}} + \mathbf{1}_{T\geq n-1} Z_{n-1} = Z_{n-1}^T. \end{aligned} \tag{3.21}$$

This, however, does not show that the stopped process is a martingale, as we need to show that $\mathbb{E}[Z_n^T \mid z_{n-1}^T, \ldots, z_1^T] = z_{n-1}^T$ for all $n \geq 1$. This is the same as $\mathbb{E}[Z_n^T \mid \mathcal{B}_{n-1}]$, where $\mathcal{B}_{n-1}$ are the events observable by $Z_1^T, \ldots, Z_{n-1}^T$, which may not be the same as $\mathcal{A}_{n-1}$. However, since $\mathcal{B}_{n-1} \subseteq \mathcal{A}_{n-1}$, we have

$$\mathbb{E}[Z_n^T \mid \mathcal{B}_{n-1}] = \mathbb{E}[\mathbb{E}[Z_n^T \mid \mathcal{A}_{n-1}] \mid \mathcal{B}_{n-1}] = Z_{n-1}^T. \tag{3.22}$$

■

Corollary 3.13. *If* T *is a stopping time for a martingale* $\{Z_n\}_{n\geq 1}$*, then* $\mathbb{E}[Z_n^T] = \mathbb{E}[Z_1]$.

Proof. From above, $\{Z_n^T\}_{n\geq 1}$ is a martingale, and hence $\mathbb{E}[Z_n^T] = \mathbb{E}[Z_1^T] = \mathbb{E}[Z_1]$. ■

Note that $Z_n^T \xrightarrow{n\to\infty} Z_T$ with probability 1, and that $\mathbb{E}[Z_n^T] = \mathbb{E}[Z_1]$. Does this imply that $\mathbb{E}[Z_n^T] \xrightarrow{n\to\infty} \mathbb{E}[Z_T]$? The following is a counterexample to this claim.

Example 3.14. Let $\{Z_n\}_{n\geq 1}$ be the product martingale sequence; that is, $\mathbb{P}(Z_n = 2^n) = \frac{1}{2^n}$ and $\mathbb{P}(Z_n = 0) = 1 - \frac{1}{2^n}$. Let $T = \min\{n \geq 1 \mid Z_n = 0\}$ be a stopping time. Then $Z_T = 0$ with probability 1, so $\mathbb{E}[Z_T] = 0$. However, $\mathbb{E}[Z_n^T] = \mathbb{E}[Z_1] = 1$. Thus, $Z_n^T \xrightarrow{n\to\infty} Z_T$ with probability 1, but $\mathbb{E}[Z_n^T]$ does not converge to $\mathbb{E}[Z_T]$.

Theorem 3.15 (The *optimal stopping theorem*)**.** *Let* T *be a stopping time for a martingale* $\{Z_n\}_{n\geq 1}$*. Then* $\mathbb{E}[Z_T] = \mathbb{E}[Z_1]$ *if and only if* $\mathbb{E}[|Z_T|] < \infty$ *and* $\mathbb{E}[Z_n \mid T \geq n] \cdot \mathbb{P}(T \geq n) \xrightarrow{n\to\infty} 0$.

Example 3.16. Consider $\{S_n\}_{n\geq 1}$ to be the simple asymmetric random walk on $\mathbb{Z}$, with step probabilities $\mathbb{P}(X_i = 1) = p = 1 - \mathbb{P}(X_i = -1)$. Then the sequence $\{W_n\}_{n\geq 1}$ defined by $W_n = \left(\frac{1-p}{p}\right)^{S_n}$ is a martingale. Now, for $a < 0 < b$, consider the stopping times $\sigma_a = \min\{n \geq 1 \mid S_n = a\}$ and $\sigma_b = \min\{n \geq 1 \mid S_n = b\}$, and let $T = \sigma_a \wedge \sigma_b$, a stopping time for $\{W_n\}_{n\geq 1}$. If the conditions for the optimal stopping theorem holds, then $\mathbb{E}[W_T] = \mathbb{E}[W_1]$. The left term is

$$\mathbb{E}[W_{\sigma_a} \mid \sigma_a < \sigma_b] + \mathbb{E}[W_{\sigma_b} \mid \sigma_a \geq \sigma_b] = \left(\frac{1-p}{p}\right)^a \mathbb{P}(\sigma_a < \sigma_b) + \left(\frac{1-p}{p}\right)^b \mathbb{P}(\sigma_a \geq \sigma_b), \tag{3.23}$$

and the right term is $\mathbb{E}[W_1] = 1$. Thus, $\mathbb{E}[W_T] = 1$. Solving this alongside $\mathbb{P}(\sigma_a < \sigma_b) + \mathbb{P}(\sigma_b \leq \sigma_a) = 1$ gives

$$\mathbb{P}(\sigma_a < \sigma_b) = \frac{\left(\frac{1-p}{p}\right)^b - 1}{\left(\frac{1-p}{p}\right)^b - \left(\frac{1-p}{p}\right)^a} \tag{3.24}$$

for $a < 0 < b$. This is the probability that the simple asymmetric random walk hits a before hitting b. We now show that the conditions for the optimal stopping theorem holds. First, we have

$$\mathbb{E}[|W_T|] = \mathbb{E}[W_T] < \infty \tag{3.25}$$

since W_T can only take finitely many values. Also, since $\mathbb{P}(T \geq n) \to 0$ as $n \to \infty$, and $\mathbb{E}[W_n \mid T \geq n]$ is bounded, we have $\mathbb{E}[W_n \mid T \geq n] \cdot \mathbb{P}(T \geq n) \to 0$ as $n \to \infty$. Hence, the conditions for the optimal stopping theorem holds.

Proof of the optimal stopping theorem. Since $\{Z_n^T\}_{n\geq 1}$ is a martingale, we have $\mathbb{E}[|Z_n^T|] < \infty$ and $\mathbb{E}[Z_n^T] = \mathbb{E}[Z_1]$ for all $n \geq 1$, and $Z_n^T \xrightarrow{n\to\infty} Z_T$ with probability 1. Now,

$$\begin{aligned}\mathbb{E}[Z_n^T] &= \sum_{i=1}^{n-1} \mathbb{E}[Z_n^T; T = i] + \mathbb{E}[Z_n^T; T \geq n] = \sum_{i=1}^{n-1} \mathbb{E}[Z_i \mid T = i] \cdot \mathbb{P}(T = i) + \mathbb{E}[Z_n \mid T \geq n] \cdot \mathbb{P}(T \geq n) \\ &= \sum_{i=1}^{n-1} \mathbb{E}[Z_i \mid T = i] \cdot \mathbb{P}(T = i) + \mathbb{E}[Z_n \mid T \geq n] \cdot \mathbb{P}(T \geq n). \end{aligned} \tag{3.26}$$

Also, $\mathbb{E}[Z_T] = \sum_{i=1}^{\infty} \mathbb{E}[Z_i \mid T = i] \cdot \mathbb{P}(T = i)$ if and only if $\mathbb{E}[|Z_T|] < \infty$. Hence, $\mathbb{E}[Z_n^T] \xrightarrow{n\to\infty} \mathbb{E}[Z_T]$ if and only if $\mathbb{E}[|Z_T|] < \infty$ and $\mathbb{E}[Z_n \mid T \geq n] \cdot \mathbb{P}(T \geq n) \xrightarrow{n\to\infty} 0$. Since $\mathbb{E}[Z_n^T] = \mathbb{E}[Z_1]$ for all n, we have $\mathbb{E}[Z_T] = \mathbb{E}[Z_1]$ if and only if the two conditions above hold. ■

The following serves as a process of checking the conditions for the optimal stopping theorem.

1. If $\mathbb{P}(T < \infty) = 1$ (that is, $\mathbb{P}(T \geq n) \to 0$ as $n \to \infty$), and $\{Z_n\}_{n\geq 1}$ is a bounded martingale.
2. $\mathbb{E}[T] < \infty$ and there exists $c > 0$ such that $|Z_n - Z_{n-1}| < c$ for all $n \geq 2$.
3. $T : \Omega \to \mathbb{N}$, and $\sup_{\omega\in\Omega} T(\omega) < \infty$.

3.3 Kolmogorov's Maximal Inequalities

October 7th.

Recall the inequalities of Markov and Chebyshev. Let X be a non-negative random variable, and let $a > 0$. Then, we have

$$\mathbb{P}(X > a) \leq \frac{\mathbb{E}[X]}{a}, \qquad \mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}. \tag{3.27}$$

Theorem 3.17 (*Kolmogorov's maximal inequality*)**.** *Let $\{Z_n\}_{n\geq 1}$ be a martingale, with $\mathbb{E}[Z_n^2] < \infty$ for all $n \geq 1$. Then, for any all $n \geq 1$ and $b > 0$, we have*

$$\mathbb{P}\left(\max_{1\leq i\leq n} |Z_i| > b\right) \leq \frac{\mathbb{E}[Z_n^2]}{b^2}. \tag{3.28}$$

If $h : \mathbb{R} \to \mathbb{R}$ is a convex function, then by Jensen's inequality, we have

$$\mathbb{E}[h(Z_n) \mid z_{n-1}, \ldots, z_1] \geq h(\mathbb{E}[Z_n \mid z_{n-1}, \ldots, z_1]) = h(z_{n-1}), \tag{3.29}$$

which is the property of a submartingale.

Definition 3.18. A sequence of random variables $\{Z_n\}_{n\geq 1}$ is a *submartingale* if $\mathbb{E}[|Z_n|] < \infty$ for all $n \geq 1$, and

$$\mathbb{E}[Z_n \mid z_{n-1}, \ldots, z_1] \geq z_{n-1} \tag{3.30}$$

As above, if $\{Z_n\}_{n\geq 1}$ is a martingale, then $\{h(Z_n)\}_{n\geq 1}$ is a submartingale for any convex function $h : \mathbb{R} \to \mathbb{R}$. In particular, $\{Z_n^2\}_{n\geq 1}$, $\{|Z_n|\}_{n\geq 1}$, and $\{|Z_n|\}_{n\geq 1}$ are submartingales.

Proposition 3.19. *Let $\{Z_n\}_{n\geq 1}$ be a non-negative submartingale. Then for any $n \geq 1$ and $a > 0$, we have*

$$\mathbb{P}\left(\max_{1\leq i\leq n} Z_i > a\right) \leq \frac{\mathbb{E}[Z_n]}{a}. \tag{3.31}$$

We shall first use this proposition to prove Kolmogorov's maximal inequality.

Proof of Kolmogorov's maximal inequality. Suppose $\{Z_n\}_{n\geq 1}$ is a martingale with all finite second moments. Then $\{Z_n^2\}_{n\geq 1}$ is a submartingale, and the above proposition gives

$$\mathbb{P}\left(\max_{1\leq i\leq n} |Z_i| > a\right) = \mathbb{P}\left(\max_{1\leq i\leq n} Z_i^2 > a^2\right) \leq \frac{\mathbb{E}[Z_n]^2}{a^2}. \tag{3.32}$$

■

Proof. Let $J = \min\{k \geq 1 \mid Z_k \geq a\} \wedge m$ for some $m \geq 1$. Then $Z_J \geq a$ holds if and only if $Z_n \geq a$ for all $n \geq m$. Then, we have

$$\begin{aligned}
\mathbb{P}\left(\max_{1\leq i\leq m} Z_i > a\right) = \mathbb{P}(Z_J \geq a) &\leq \frac{\mathbb{E}[Z_J]}{a} = \frac{1}{a}\sum_{k=1}^{m-1} \mathbb{E}[Z_k \mathbf{1}_{\{J=k\}}] + \frac{1}{a}\mathbb{E}[Z_m \mathbf{1}_{\{J=m\}}] \\
&= \frac{1}{a}\sum_{k=1}^{m-1} \mathbb{E}[\mathbb{E}[Z_m \mathbf{1}_{\{J=k\}} \mid \mathcal{A}_k]] + \frac{1}{a}\mathbb{E}[Z_m \mathbf{1}_{\{J=m\}}] \\
&= \frac{1}{a}\sum_{k=1}^{m-1} \mathbb{E}[\mathbf{1}_{\{J=k\}}\mathbb{E}[Z_m \mid \mathcal{A}_k]] + \frac{1}{a}\mathbb{E}[Z_m \mathbf{1}_{\{J=m\}}] \\
&\leq \frac{1}{a}\sum_{k=1}^{m-1} \mathbb{E}[\mathbf{1}_{\{J=k\}} Z_k] + \frac{1}{a}\mathbb{E}[Z_m \mathbf{1}_{\{J=m\}}] = \frac{1}{a}\sum_{k=1}^{m} \mathbb{E}[Z_k \mathbf{1}_{\{J=k\}}] = \frac{\mathbb{E}[Z_m]}{a}.
\end{aligned} \tag{3.33}$$

■

Proposition 3.20. *Let $\{Z_n\}_{n\geq 1}$ be a non-negative martingale. Then*

$$\mathbb{P}\left(\sup_{n\geq 1} Z_n > a\right) \leq \frac{\mathbb{E}[Z_1]}{a} \tag{3.34}$$

for all $a > 0$.

Proof. For $m \geq 1$, define $A_m = \{\max_{1\leq i\leq m} Z_i > a\} = \bigcup_{i=1}^m \{Z_i > a\}$. Then, $A_m \subseteq A_{m+1}$; but note that $\bigcup_{m=1}^\infty A_m = \bigcup_{n=1}^\infty \{Z_n > a\}$ is *not* the same as $\{\sup_{n\geq 1} Z_n > a\}$, as the latter is the event that $Z_n > a$ for infinitely many n, and the former is the event that $Z_n > a$ for some n. Let $\varepsilon_0 \in (0, a)$. Then, we have

$$\mathbb{P}\left(\sup_{i\geq 1} Z_i \geq a\right) = \mathbb{P}\left(\bigcap_{\varepsilon>0}\bigcup_{i\geq 1}\{Z_i \geq a - \varepsilon\}\right) \leq \mathbb{P}(\bigcup_{i\geq 1}\{Z_i \geq a - \varepsilon_0\}). \tag{3.35}$$

Let $b = a - \varepsilon_0$. By the previous proposition, we have

$$\mathbb{P}\left(\max_{1\leq i\leq m} Z_i \geq b\right) \leq \frac{\mathbb{E}[Z_1]}{b}. \tag{3.36}$$

Taking the limit as $m \to \infty$ gives

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty}\{Z_i \geq b\}\right) \leq \frac{\mathbb{E}[Z_1]}{b}. \tag{3.37}$$

Combining the above inequalities gives

$$\mathbb{P}\left(\sup_{i\geq 1} Z_i \geq a\right) \leq \frac{\mathbb{E}[Z_1]}{b} = \frac{\mathbb{E}[Z_1]}{a-\varepsilon_0}. \tag{3.38}$$

Letting $\varepsilon_0 \to 0$ gives the desired result. ■

Theorem 3.21 (Convergence theorem)**.** *Let $\{Z_n\}_{n\geq 1}$ be a martingale.*

1. *If $Z_n \geq 0$ for all $n \geq 1$, then there exists a random variable Z such that $Z_n \xrightarrow{n\to\infty} Z$ with probability 1.*
2. *If $\sup_{n\geq 1}\mathbb{E}[|Z_n|] < \infty$, then there exists a random variable Z such that $Z_n \xrightarrow{n\to\infty} Z$ with probability 1.*
3. *If $\sup_{n\geq 1}\mathbb{E}[Z_n^2] < \infty$, then there exists a random variable Z such that $Z_n \xrightarrow{n\to\infty} Z$ with probability 1.*

Proof. Note that 2. implies 1. since if $Z_n \geq 0$, then $\mathbb{E}[|Z_n|] = \mathbb{E}[Z_n] = \mathbb{E}[Z_1]$ implies that $\sup_{n\geq 1}\mathbb{E}[|Z_n|] < \infty$. Also, 2. implies 3. since $\mathbb{E}[|Z_n|]^2 \leq \mathbb{E}[Z_n^2]$ by Jensen's inequality. For a proof of 3., note that $\sup_{n\geq 1}\mathbb{E}[Z_n^2] < \infty$ implies that $\{Z_n^2\}_{n\geq 1}$ is a submartingale with bounded expectation. Taking expectation on $\mathbb{E}[Z_n^2 \mid Z_{n-1}, \ldots, Z_1] \geq \mathbb{E}[Z_{n-1}^2]$ gives $\mathbb{E}[Z_n^2] \geq \mathbb{E}[Z_{n-1}^2]$, and hence $\mathbb{E}[Z_n^2]$ is non-decreasing. Since it is bounded, we have $\mathbb{E}[Z_n^2] \xrightarrow{n\to\infty} \alpha$ for some $0 \leq \alpha < \infty$. Now fix $m \geq 1$, and set $Y_n = Z_{n+m} - Z_m$ for $n \geq 1$. Then, $\{Y_n\}_{n\geq 1}$ is a martingale, and Kolmogorov's maximal inequality gives

$$\mathbb{P}\left(\max_{1\leq i\leq n}|Y_i| > b\right) \leq \frac{\mathbb{E}[Y_n^2]}{b^2} \tag{3.39}$$

for all $b > 0$, and also $\mathbb{E}[Y_n^2] = \mathbb{E}[Z_{n+m}^2] - 2\mathbb{E}[Z_{n+m}Z_m] + \mathbb{E}[Z_m^2]$. Since $\{Z_n\}_{n\geq 1}$ is a martingale, we have $\mathbb{E}[Z_{n+m} \mid Z_m, \ldots, Z_1] = Z_m$, and hence $\mathbb{E}[Z_{n+m}Z_m] = \mathbb{E}[\mathbb{E}[Z_{n+m}Z_m \mid Z_m, \ldots, Z_1]] = \mathbb{E}[Z_m^2]$. Thus, $\mathbb{E}[Y_n^2] = \mathbb{E}[Z_{n+m}^2] - \mathbb{E}[Z_m^2]$, which is bounded by $\alpha - \mathbb{E}[Z_m^2]$ as $n \to \infty$. So,

$$\mathbb{P}\left(\max_{1\leq i\leq n}|Z_{i+m} - Z_i| > b\right) \leq \frac{\mathbb{E}[Z_{n+m}^2] - \mathbb{E}[Z_m^2]}{b^2} \tag{3.40}$$

for all $n, m \geq 1$. Letting $n \to \infty$ on both sides gives

$$\mathbb{P}\left(\bigcup_{i\geq 1}\{|Z_{i+m} - Z_i| > b\}\right) \leq \frac{\alpha - \mathbb{E}[Z_m^2]}{b^2}. \tag{3.41}$$

Now letting $m \to \infty$ gives

$$\lim_{m\to\infty}\mathbb{P}\left(\bigcup_{i\geq 1}\{|Z_{i+m} - Z_i| > b\}\right) = 0 \qquad \text{for all } b > 0. \tag{3.42}$$

We claim that this implies that $\{Z_n\}_{n\geq 1}$ is a Cauchy sequence with probability 1. Indeed, look at

$$A = \bigcap_{b>0}\bigcup_{N=1}^{\infty}\bigcap_{m=N+1}^{\infty}\bigcap_{i=N+1}^{\infty}\{|Z_{i+m} - Z_i| < b\}. \tag{3.43}$$

Then, A is the event that $\{Z_n\}_{n\geq 1}$ is a Cauchy sequence; it is enough to show that $\mathbb{P}(A) = 1$. Let $l \geq 1$ and set $b = \frac{1}{2^l}$. By the above, there exists k_l such that

$$\mathbb{P}\left(\bigcup_{i\geq 1}|Z_{i+k_l} - Z_i| > \frac{1}{2^l}\right) \leq \frac{1}{2^l}. \tag{3.44}$$

This implies

$$\sum_{l\geq 1}\mathbb{P}\left(\bigcup_{i\geq 1}|Z_{i+k_l}-Z_i|>\frac{1}{2^l}\right)<\infty \implies \sum_{l\geq 1}\mathbb{P}\left(|Z_{i+k_l}-Z_i|\leq\frac{1}{2^l}\right)<\infty. \tag{3.45}$$

By the Borel-Cantelli lemma, we receive

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty}\bigcup_{m=n}^{\infty}A_m\right)=0. \tag{3.46}$$

Taking complements gives $\mathbb{P}(A)=1$, and hence $\{Z_n\}_{n\geq 1}$ is a Cauchy sequence with probability 1. Since $\mathbb{R}$ is complete, there exists a random variable Z such that $Z_n \xrightarrow{n\to\infty} Z$ with probability 1.

Now, for a proof of 2., for all $a,b\in\mathbb{Q}$, define $\bigwedge_{a,b}=\{\omega\in\Omega \mid \liminf_{n\to\infty}Z_n<a<b<\limsup_{n\to\infty}Z_n\}$ where $(Z_n)_{n\geq 1}$ is a martingale. Then the probability that $\{Z_n\}_{n\geq 1}$ does not converge and does not properly diverge is $\mathbb{P}(\bigcup_{a,b\in\mathbb{Q}}\bigwedge_{a,b})$. By subadditivity of the probability measure, we have

$$\mathbb{P}\left(\bigcup_{a,b\in\mathbb{Q}}\bigwedge_{a,b}\right)\leq\sum_{a,b\in\mathbb{Q}}\mathbb{P}\left(\bigwedge_{a,b}\right). \tag{3.47}$$

We claim that $\mathbb{P}(\bigwedge_{a,b})=0$ for all $a,b\in\mathbb{Q}$. To this end, let fix $a,b\in\mathbb{Q}$. Also set $S_1=\min\{n\geq 1 \mid Z_n<a\}$, $t_n=\min\{k\geq S_{n-1}\mid Z_k>b\}$, and $S_n=\min\{k\geq t_{n-1}\mid Z_k<a\}$. Now define $U_N(a,b)=\max\{k\geq 1 \mid S_1<t_1<S_2<t_2<\ldots<S_k<t_k\leq N\}$, the number of upcrossings of the interval (a,b) by time N. Then, note that $U_N(a,b)\to U_{a,b}$ as $N\to\infty$; the event $\bigwedge_{a,b}=\{U_{a,b}=\infty\}$. So it is enough to show that $\mathbb{P}(U_{a,b}=\infty)=0$. Now, set $V_1=\mathbf{1}_{\{Z_1<a\}}$ and $V_n=1_{\{V_{n-1}=1,Z_{n-1}\leq b\}}+\mathbf{1}_{\{V_{n-1}=0,Z_{n-1}<a\}}$ for $n\geq 2$. Then set $Z_N^V=\sum_{k=1}^N V_k(Z_k-Z_{k-1})$. Then, $\{Z_N^V\}_{N\geq 1}$ is a martingale, and $Z_N^V\leq(b-a)U_N(a,b)+(Z_N-a)^+$. Taking expectations gives

$$\mathbb{E}[U_N(a,b)]\leq\frac{1}{b-a}(\mathbb{E}[Z_N^V]+\mathbb{E}[(Z_N-a)^-])\leq\frac{1}{b-a}(\mathbb{E}[Z_1]+\mathbb{E}[|Z_n|]+a)\leq c. \tag{3.48}$$

[proof incomplete]

Thus, the probability that $\{Z_n\}_{n\geq 1}$ does not converge and does not properly diverge is 0. However, using Fatou's lemma, we know that

$$\mathbb{E}[|Z|]=\mathbb{E}[\liminf_{n\to\infty}|Z_n|]\leq\liminf_{n\to\infty}\mathbb{E}[|Z_n|]<\infty \tag{3.49}$$

or that $\mathbb{P}(Z\in\mathbb{R})=1$. Hence, $\{Z_n\}_{n\geq 1}$ converges with probability 1. ■

Chapter 4

APPLICATIONS

October 14th.

Example 4.1. A monkey randomly chooses an alphabet key on a typewriter uniformly; let u_i be the key chosen at i-th time. What is the chance the monkey types out "ISP IS BORING"?

In particular, let

$$\tau = \min\{n \geq 11 \mid (u_{n-10}, \dots, u_n) = (I, S, P, I, S, B, O, R, I, N, G)\}. \tag{4.1}$$

We ask if $\mathbb{E}[\tau]$ is finite. The Markov chain method is as follows; for all $k \geq 1$, define $Y_k = (u_k, \dots, u_{k+10})$. As $u_i \sim \text{Unif}\{A, B, \dots, Z\}$, $\{Y_k\}_{k\geq 1}$ turns into a Markov chain. Using $h(x) = \mathbb{E}[\tau \mid Y_1 = x]$, we may set up a recurrence relation and solve for h by hand to obtain an answer. This is computationally expensive by hand.

A better approach is to use martingales. Let $(Y_1, \dots, Y_{11}) = (I, S, P, \dots, I, N, G)$, and set $Z_0^1 = 1$ to be the capital of gambler 1 at time $t = 0$. At time $t = 1$, the gambler places a bet that $u_1 = I$. If $u_1 = I$, then the gambler gets 26 capital, and if $u_1 \neq I$, the gambler gets 0 capital and exits the casino. The expected gain here is then

$$\frac{1}{26}(26 - 1) + \frac{25}{26}(-1) = 0. \tag{4.2}$$

At time $t = 2$, the gambler bets all the gain, 26 capital, that $u_2 = S$. If $u_2 = S$, the gambler gets $(26)^2$ capital, and if $u_2 \neq S$, the gambler loses and exits the casino. The expected gain here is, again, 0. Repeat this until $t = 11$, and $Z_t^1 = Z_{11}^1$ for all $t \geq 10$, where Z_n^1 is the current capital; that is,

$$Z_n^1 = \begin{cases} 26 \cdot Z_{n-1}^1 \mathbf{1}_{\{u_n = \psi_n\}} & \text{if } 1 \leq n \leq 11, \\ Z_{11}^1 & \text{if } n \geq 12. \end{cases} \tag{4.3}$$

Now, a second gambler enters the casino at time $t = 2$ and places a bet of 1 that $u_2 = I$. If $u_2 = I$, the second gambler gets 26 capital, and 0 otherwise. Continue similarly till $t = 12$. So, for any $k \geq 1$, gambler k enters at time $t = k$ and starts with capital 1 for bet $u_k = I$, with capital at each step as

$$Z_n^k = \begin{cases} 1 & \text{if } n < k, \\ 26 \cdot Z_{n-1}^k \mathbf{1}_{\{u_n = \psi_{n-k+1}\}} & \text{if } k \leq n \leq k + 10, \\ Z_{k+10}^k & \text{if } n \geq k + 11. \end{cases} \tag{4.4}$$

Now, finally, define

$$M_n = \sum_{k=1}^{n} (1 - Z_n^k) = n - \sum_{k=1}^{n} Z_n^k. \tag{4.5}$$

the gain of the casino. We make three claims:

1. $(M_n)_{n\geq 1}$ is a martingale.
2. τ is a stopping time for this martingale.
3. The OST applies.

To show $(M_n)_{n\geq 1}$ is a martingale, we first have

$$|M_n| \leq n + n(26)^{11} < \infty, \tag{4.6}$$

so they are integrable. Moreover,

$$\begin{aligned}
\mathbb{E}[M_n \mid M_{n-1}, \dots, M_1] &= n - \mathbb{E}\left[\sum_{k=1}^{n-11} Z_{n-1}^k + \sum_{k=n-10}^{n} 26 \cdot Z_{n-1}^k \mathbf{1}_{\{u_n = \psi_{n-k+1}\}} \mid u_{n-1}, \dots, u_1\right] \\
&= n - \sum_{k=1}^{n-11} Z_{n-1}^k + 26 \sum_{k=n-10}^{n} Z_{n-1}^k \mathbb{P}(u_n = \psi_{n-k+1}) \\
&= n - \sum_{k=1}^{n-11} Z_{n-1}^k - \sum_{k=n-10}^{n} Z_{n-1}^k = n - 1 - \sum_{k=1}^{n-1} Z_{n-1}^k = M_{n-1},
\end{aligned} \tag{4.7}$$

showing that $(M_n)_{n\geq 1}$ is a martingale. For the second claim, note that $\{\tau \leq n\} = \bigcup_{k=1}^{n} \{(u_k, \dots, u_{k+10}) = (I, S, P, \dots, I, N, G)\} \in \sigma(u_1, \dots, u_n)$, so τ is a stopping time. Now, $M_n = n - \sum_{k=1}^{n} Z_n^k$, and $\mathbb{E}[M_1] = 1 - \mathbb{E}[Z_1^1] = 0$. Note that $Z_\tau^k = 0$ for all $k \leq \tau - 10$, and $Z_\tau^k = 0$ for all $\tau - 10 < k \leq \tau$ except for $k = \tau - 10$, where $Z_\tau^{\tau-10} = 26^{11}$. If the OST applies, then $\mathbb{E}[M_\tau] = \mathbb{E}[M_1] = 0$, so

$$\mathbb{E}[\tau] = \mathbb{E}[M_\tau] + (26)^{11} = (26)^{11} < \infty. \tag{4.8}$$

Thus, we are only left to show that the OST applies. If we let W_k denote the k-th 11-block of u's, then the W_k's are independent and identically distributed. If S is the first k such that $W_k = (I, S, P, \dots, I, N, G)$, then $\tau \leq 11S$. But this implies

$$\mathbb{E}[\tau] \leq 11\mathbb{E}[S] = 11 \cdot 26^{11} < \infty, \tag{4.9}$$

so τ is integrable. Now, the increments in M_n are handled as

$$\begin{aligned}
|M_n - M_{n-1}| &= \left|\sum_{k=1}^{n-1} (Z_{n-1}^k - Z_n^k) - Z_n^n + 1\right| \leq \left|\sum_{k=1}^{n-1} (Z_{n-1}^k - Z_n^k)\right| + 25 \\
&\leq 25 \sum_{k=n-10}^{n-1} \left|Z_{n-1}^k\right| + 25 \leq 25(26)^{11} 10 + 25.
\end{aligned} \tag{4.10}$$

The OST applies, and we are done.

Example 4.2. Let S_n be the permutation group on n elements, $S_n = \{\sigma : \{1, \dots, n\} \to \{1, \dots, n\} \mid \sigma \text{ is bijective}\}$. Let $\pi \in S_n$. An increasing subsequence of π is a sequence $1 \leq i_1 < i_2 < \dots < i_k \leq n$ such that $\pi(i_1) < \pi(i_2) < \dots < \pi(i_k)$. Let $L_n(\pi)$ denote the length of the longest increasing subsequence of π, where π is chosen uniformly at random from S_n. What is the asymptotic behavior of $L_n = L_n(\pi)$ as $n \to \infty$?

We show the following theorem.

Theorem 4.3. *The following hold for L_n.*

1. $\mathbb{E}[L_n] \geq \sqrt{n}$.
2. $\limsup_{n\to\infty} \frac{\mathbb{E}[L_n]}{\sqrt{n}} \leq e$.
3. $\mathbb{P}(|L_n - \mathbb{E}[L_n]| \geq t\sqrt{n}) \leq 2e^{-t^2/2}$.

From these statements, one can conclude that

$$1 \leq \liminf_{n\to\infty} \frac{\mathbb{E}[L_n]}{\sqrt{n}} \leq \limsup_{n\to\infty} \frac{\mathbb{E}[L_n]}{\sqrt{n}} \leq e, \tag{4.11}$$

and

$$\frac{L_n}{\sqrt{n}} - \frac{\mathbb{E}[L_n]}{\sqrt{n}} \xrightarrow{\mathrm{P}} 0. \tag{4.12}$$

We make use of the following fact, that if $x_1, \ldots, x_n$ are real numbers $L(x_1, \ldots, x_n)$ and $D(x_1, \ldots, x_n)$ denoting the lenghts of the longest increasing and decreasing subsequences of $x_1, \ldots, x_n$, then

$$L(x_1, \ldots, x_n)D(x_1, \ldots, x_n) \geq n. \tag{4.13}$$

Note that $D_n(\pi)$, the length of the longest decreasing subsequence of π, is equal in distribution to $L_n(\pi)$, so $\mathbb{E}[L_n]^2 \geq n$, giving the first statement. For the second statement, let N_k denote the number of increasing subsequences of length k. Then

$$N_k = \sum_{1\leq i_1<\ldots<i_k\leq n} \mathbf{1}_{\{\pi(i_1)<\ldots<\pi(i_k)\}}. \tag{4.14}$$

Then $\mathbb{E}[N_k] = \binom{n}{k}\frac{1}{k!}$. Then using Markov's inequality and Stirling's formula gives

$$\mathbb{P}(L_n \geq k) = \mathbb{P}(N_k \geq 1) \leq \mathbb{E}[N_k] = \binom{n}{k}\frac{1}{k!} \implies \mathbb{E}[L_n] = \sum_{k=1}^{n} \mathbb{P}(L_n \geq k) \leq \sum_{k=1}^{\infty} \binom{n}{k}\frac{1}{k!} \leq \sqrt{n}e. \tag{4.15}$$

For the last statement, let $\mathcal{A}_k = \sigma(\pi(1), \ldots, \pi(k))$, and set $M_k = \mathbb{E}[L_n \mid \mathcal{A}_k]$ for $n \geq 1$, and $M_0 = \mathbb{E}[L_n]$. Then $(M_k)_{0\leq k\leq n}$ is a martingale (show this). Let $D_k = M_k - M_{k-1}$. Then

$$\mathbb{E}[D_k \mid \mathcal{A}_{k-1}] = \mathbb{E}[M_k \mid \mathcal{A}_{k-1}] - M_{k-1} = 0. \tag{4.16}$$

Also, there exists c_k such that $|D_k| \leq c_k$ (show this). This implies that

$$\mathbb{E}[e^{\lambda D_k} \mid \mathcal{A}_{k-1}] \leq e^{\lambda^2 c_k^2/2} \tag{4.17}$$

or

$$\mathbb{E}[e^{\lambda(M_n - M_0)}] = \mathbb{E}[e^{\lambda\sum_{k=1}^{n} D_k}] = \mathbb{E}[e^{\sum_{k=1}^{n-1} D_k}\mathbb{E}[e^{\lambda D_n} \mid \mathcal{A}_{n-1}]] \leq e^{\lambda^2\sum_{k=1}^{n} c_k^2/2}. \tag{4.18}$$

Using this, we have

$$\mathbb{P}(M_n - M_0 \geq t) \leq \mathbb{P}(e^{\lambda(M_n-M_0)} \geq e^{\lambda t}) \leq e^{-\lambda t}\mathbb{E}[e^{\lambda(M_n-M_0)}] \leq e^{-\lambda t+\lambda^2\sum_{k=1}^{n} c_k^2/2}. \tag{4.19}$$

One can show that $c_k = 1$ works, so that

$$\mathbb{P}(M_n - M_0 \geq t) \leq e^{-\lambda t+n\lambda^2/2}. \tag{4.20}$$

Minimizing the right-hand side over λ gives $\lambda = t/n$, so

$$\mathbb{P}(M_n - M_0 \geq t) \leq e^{-t^2/2n}. \tag{4.21}$$

This gives the desired result, since $\sqrt{n}(M_n - M_0) = L_n - \mathbb{E}[L_n]$.

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